

Exponentials

Worked Solutions with TikZ Graphs

Wisest Maths — Year 1 A-Level, Exponentials and Logarithms (ref **e11**)

This document contains all 70 *Exponentials* questions, each with a fully worked solution and a TikZ graph of the relevant exponential function: the curve(s) plotted with the y -intercept and key points marked, the horizontal asymptote shown dashed, a dashed **tangent** for the gradient questions, and a dashed reference line through the solution for the equation-solving questions.

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Exponentials 01 (el1-001)

Difficulty: Foundation / Marks: 2

State two key features that all graphs of the form $y = a^x$ (where $a > 1$) have in common.

Worked solution.

Step 1: All exponential graphs pass through the point $(0, 1)$.

$$a^0 = 1 \text{ for any } a > 0$$

Any positive number raised to the power zero equals 1.

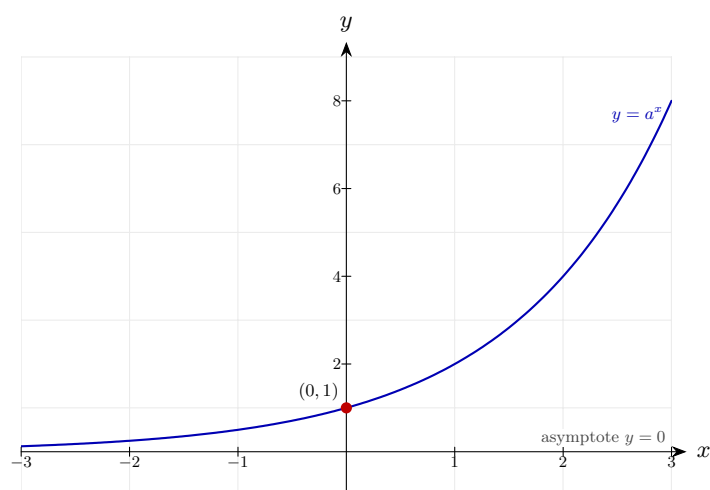
Step 2: The graph has a horizontal asymptote at $y = 0$.

$$\text{As } x \rightarrow -\infty, \quad y \rightarrow 0$$

The curve gets closer and closer to the x -axis but never touches it.

Final answer: (1) All pass through $(0, 1)$. (2) The x -axis ($y = 0$) is an asymptote.

Diagram.



The exponential function plotted with key points marked.

Exponentials 02 (e11-002)

Difficulty: Foundation / Marks: 2

Describe the difference in shape between the graph of $y = 4^x$ and the graph of $y = 2^x$ for $x > 0$.

Worked solution.

Step 1: Both graphs are increasing and pass through $(0, 1)$.

$$4^0 = 1 \quad \text{and} \quad 2^0 = 1$$

They share the same y -intercept.

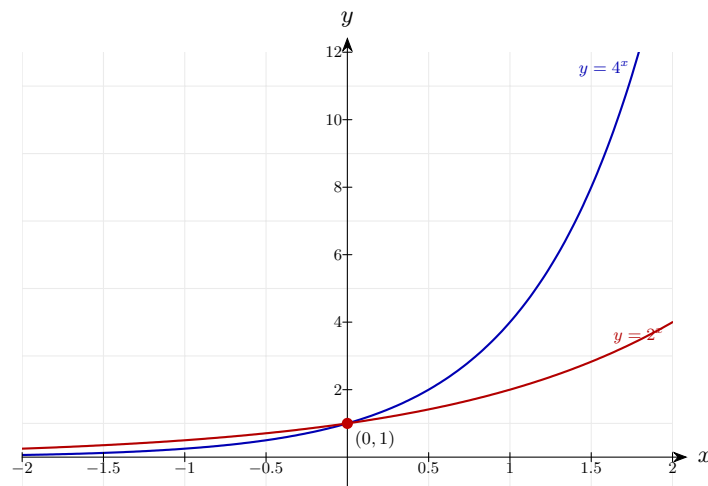
Step 2: Since $4 > 2$, the graph of $y = 4^x$ grows more steeply (faster) than $y = 2^x$ for $x > 0$.

$$4^x > 2^x \quad \text{for all } x > 0$$

A larger base means a faster rate of increase.

Final answer: Both pass through $(0, 1)$ and are increasing, but $y = 4^x$ is steeper (grows faster) than $y = 2^x$ for $x > 0$.

Diagram.



The exponential function plotted with key points marked.

Exponentials 03 (e11-003)

Difficulty: Foundation / Marks: 3

Describe the graph of $y = \left(\frac{1}{3}\right)^x$. In particular, state its y -intercept, whether it is increasing or decreasing, and the equation of its asymptote.

Worked solution.

Step 1: Find the y -intercept by setting $x = 0$.

$$\left(\frac{1}{3}\right)^0 = 1$$

y -intercept is $(0, 1)$.

Step 2: Since the base $\frac{1}{3} < 1$, the function is decreasing.

As x increases, $\left(\frac{1}{3}\right)^x$ decreases towards 0

Raising a fraction to a larger power makes it smaller.

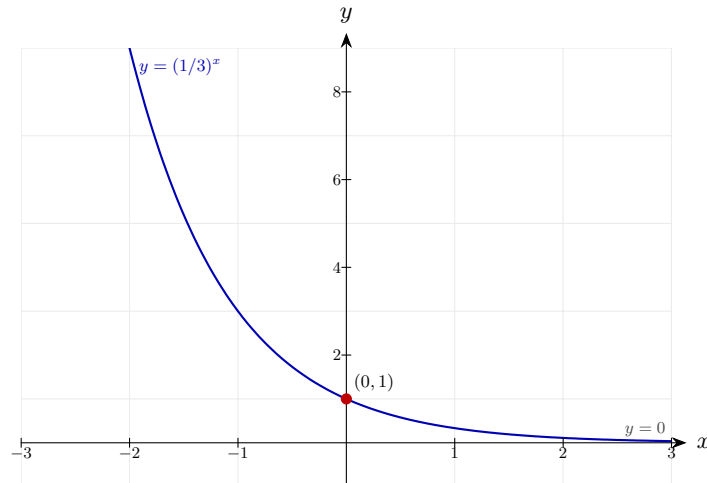
Step 3: The asymptote is the x -axis.

$$\text{As } x \rightarrow \infty, \quad y \rightarrow 0$$

The graph approaches but never reaches $y = 0$.

Final answer: y -intercept $(0, 1)$; decreasing; asymptote $y = 0$.

Diagram.



The exponential function plotted with key points marked.

Exponentials 04 (e11-004)

Difficulty: Foundation / Marks: 2

Explain why $y = \left(\frac{1}{5}\right)^x$ can also be written as $y = 5^{-x}$.

Worked solution.

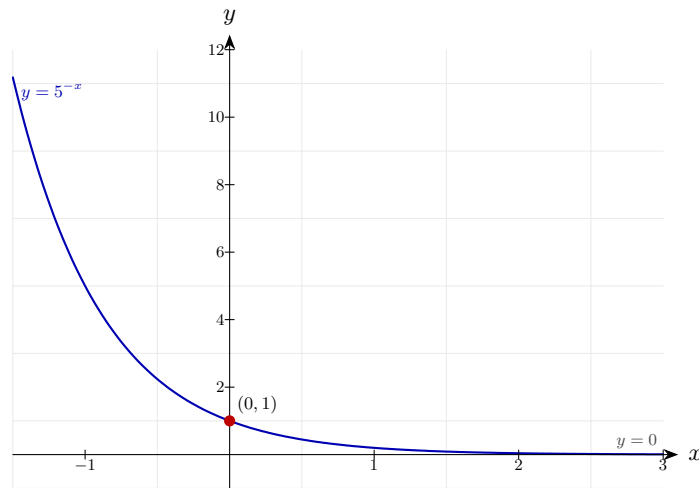
Use the rule for negative exponents.

$$\left(\frac{1}{5}\right)^x = \frac{1}{5^x} = 5^{-x}$$

$\frac{1}{a^x} = a^{-x}$ by the laws of indices.

Final answer: $\left(\frac{1}{5}\right)^x = 5^{-x}$ because raising the reciprocal to the power x is the same as raising 5 to the power $-x$.

Diagram.



The exponential function plotted with key points marked.

Exponentials 05 (e11-005)

Difficulty: Foundation / Marks: 3

Describe the transformation that maps $y = 3^x$ onto $y = 3^{-x}$. What does this tell you about the relationship between the graphs of $y = 3^x$ and $y = \left(\frac{1}{3}\right)^x$?

Worked solution.

Step 1: Replacing x with $-x$ reflects the graph in the y -axis.

$$y = 3^{-x} \text{ is a reflection of } y = 3^x \text{ in the } y\text{-axis}$$

This is the standard horizontal reflection transformation.

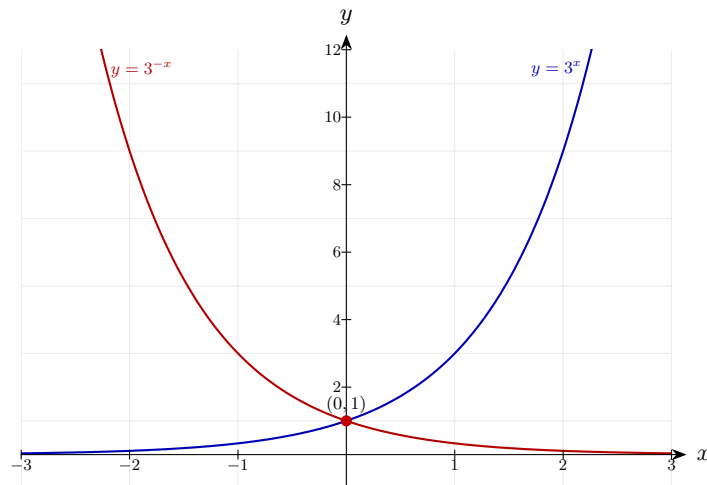
Step 2: Note that $3^{-x} = \left(\frac{1}{3}\right)^x$.

$$3^{-x} = \frac{1}{3^x} = \left(\frac{1}{3}\right)^x$$

So the two graphs are identical.

Final answer: Reflection in the y -axis. The graphs of $y = 3^x$ and $y = \left(\frac{1}{3}\right)^x$ are reflections of each other in the y -axis.

Diagram.



The exponential function plotted with key points marked.

Exponentials 06 (e11-006)

Difficulty: Foundation / Marks: 4

For the function $f(x) = 2^x$, evaluate $f(0)$, $f(3)$, $f(-2)$, and $f(\frac{1}{2})$. Give exact answers.

Worked solution.

Step 1: Evaluate $f(0)$.

$$f(0) = 2^0 = 1$$

Any base to the power zero is 1.

Step 2: Evaluate $f(3)$.

$$f(3) = 2^3 = 8$$

$$2 \times 2 \times 2 = 8.$$

Step 3: Evaluate $f(-2)$.

$$f(-2) = 2^{-2} = \frac{1}{4}$$

$$2^{-2} = \frac{1}{2^2} = \frac{1}{4}.$$

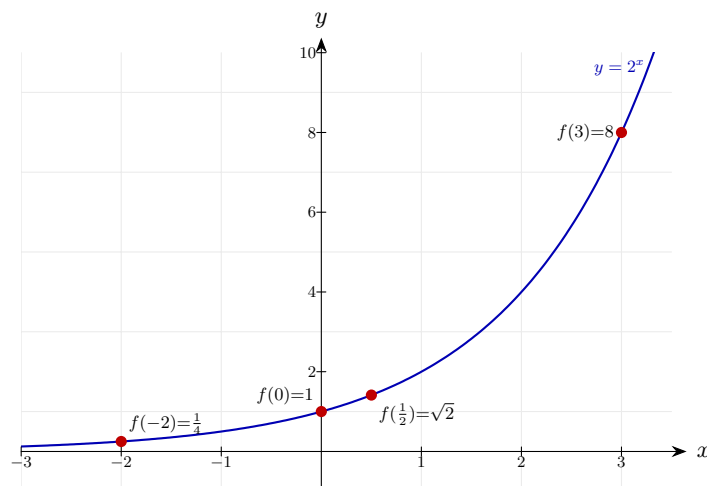
Step 4: Evaluate $f(\frac{1}{2})$.

$$f\left(\frac{1}{2}\right) = 2^{\frac{1}{2}} = \sqrt{2}$$

A power of $\frac{1}{2}$ means square root.

Final answer: $f(0) = 1$, $f(3) = 8$, $f(-2) = \frac{1}{4}$, $f\left(\frac{1}{2}\right) = \sqrt{2}$

Diagram.



The exponential function plotted with key points marked.

Exponentials 07 (e11-007)

Difficulty: Foundation / Marks: 4

On the same axes, sketch the graphs of $y = 5^x$ and $y = 5^{-x}$ for $-3 \leq x \leq 3$. Label the y -intercept and asymptote of each graph.

Worked solution.

Step 1: Both graphs pass through $(0, 1)$.

$$5^0 = 1 \quad \text{and} \quad 5^{-0} = 1$$

Common y -intercept at $(0, 1)$.

Step 2: Sketch $y = 5^x$: increasing, steeply rising for $x > 0$, approaching $y = 0$ as $x \rightarrow -\infty$.

Asymptote: $y = 0$ (left side)

Standard increasing exponential curve.

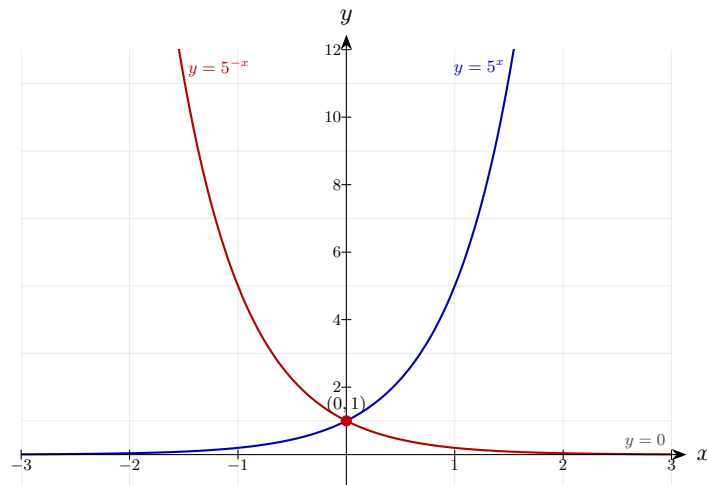
Step 3: Sketch $y = 5^{-x}$: decreasing, mirror image of $y = 5^x$ in the y -axis.

Asymptote: $y = 0$ (right side)

The two graphs are reflections of each other in the y -axis.

Final answer: Both curves pass through $(0, 1)$ with asymptote $y = 0$. $y = 5^x$ increases; $y = 5^{-x}$ decreases (its mirror image).

Diagram.



The exponential function plotted with key points marked.

Exponentials 08 (e11-008)

Difficulty: Foundation / Marks: 3

The function $f(x) = 1.8^x$.

(a) State the coordinates of the point where the graph crosses the y -axis.

(b) Is the graph increasing or decreasing?

(c) Sketch the graph.

Worked solution.

Step 1: Part (a): Set $x = 0$.

$$f(0) = 1.8^0 = 1 \Rightarrow (0, 1)$$

All exponential graphs of the form a^x cross the y -axis at 1.

Step 2: Part (b): Since the base $1.8 > 1$, the graph is increasing.

$$a = 1.8 > 1 \Rightarrow \text{increasing}$$

As x increases, 1.8^x grows.

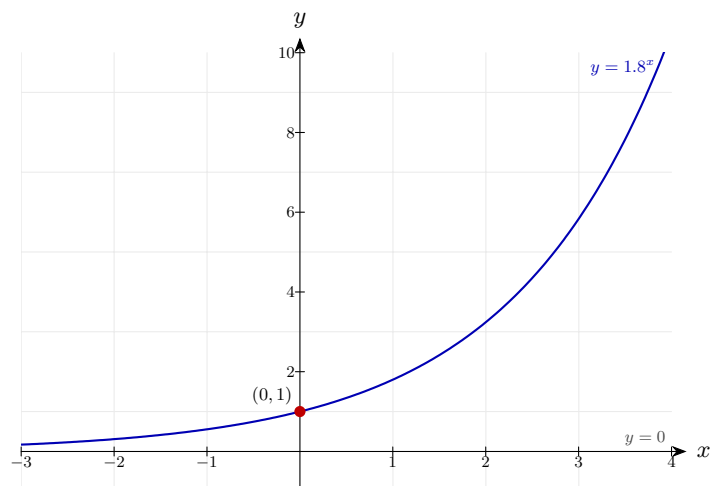
Step 3: Part (c): Sketch a smooth upward curve through $(0, 1)$, rising to the right and approaching $y = 0$ to the left.

$$\text{Asymptote: } y = 0; \quad \text{passes through } (-1, \frac{1}{1.8}) \text{ and } (1, 1.8)$$

Typical shape of $y = a^x$ for $a > 1$.

Final answer: (a) $(0, 1)$. (b) Increasing. (c) Smooth upward curve with asymptote $y = 0$.

Diagram.



The exponential function plotted with key points marked.

Exponentials 09 (e11-009)

Difficulty: Foundation / Marks: 2

State the approximate value of e and explain what is special about the function $y = e^x$ in terms of its gradient.

Worked solution.

Step 1: The value of e .

$$e \approx 2.718$$

e is an irrational number approximately equal to 2.718.

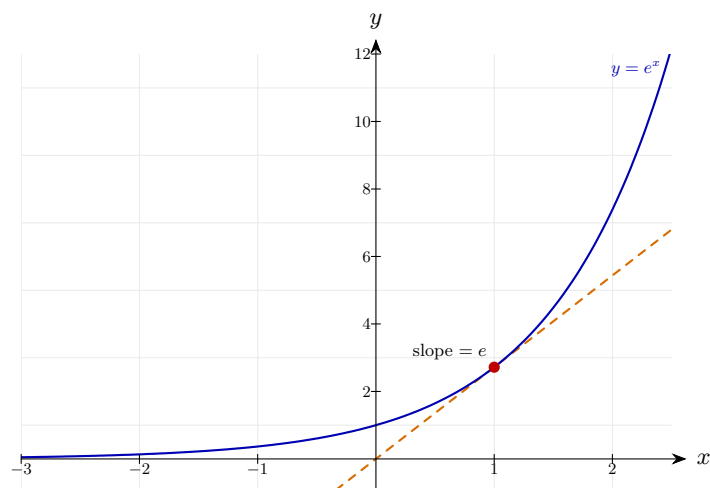
Step 2: The special gradient property.

$$\frac{d}{dx}(e^x) = e^x$$

For the function $y = e^x$, the gradient (derivative) at every point equals the value of the function itself. This is unique to e .

Final answer: $e \approx 2.718$. The gradient of $y = e^x$ equals e^x at every point — i.e., its own value.

Diagram.



The exponential function plotted with key points marked and the tangent (dashed) showing the gradient.

Exponentials 10 (e11-010)

Difficulty: Foundation / Marks: 3

Find the gradient of $y = e^{3x}$ at the point where $x = 0$. Give an exact answer.

Worked solution.

Step 1: Identify the form $y = e^{kx}$ with $k = 3$.

$$y = e^{3x}, \quad k = 3$$

This is in the standard form $y = e^{kx}$.

Step 2: The gradient of $y = e^{kx}$ is ke^{kx} .

$$\frac{dy}{dx} = 3e^{3x}$$

Multiply the function by k to get the gradient.

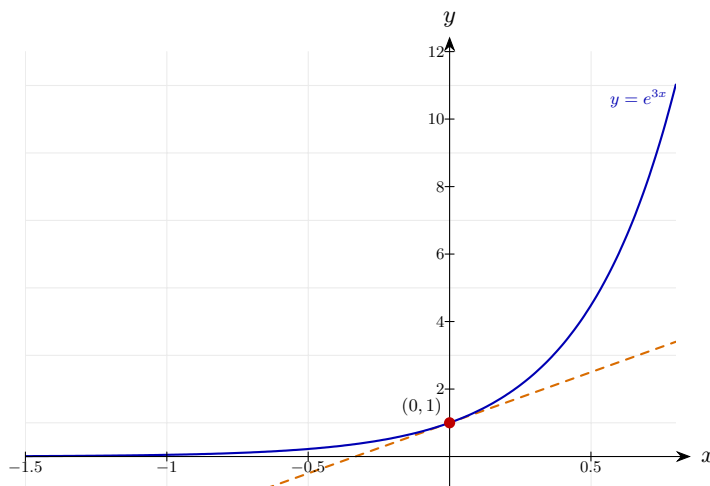
Step 3: Evaluate at $x = 0$.

$$3e^{3 \times 0} = 3e^0 = 3 \times 1 = 3$$

$$e^0 = 1.$$

Final answer: Gradient = 3

Diagram.



The exponential function plotted with key points marked and the tangent (dashed) showing the gradient.

Exponentials 11 (e11-011)

Difficulty: Foundation / Marks: 4

Find the gradient of $y = 4e^{2x}$ at the points $x = 0$ and $x = 1$. Give exact answers.

Worked solution.

Step 1: Identify $A = 4$ and $k = 2$.

$$y = 4e^{2x}, \quad A = 4, \quad k = 2$$

In the form $y = Ae^{kx}$.

Step 2: Write the gradient formula.

$$\frac{dy}{dx} = kAe^{kx} = 2 \times 4 \times e^{2x} = 8e^{2x}$$

Multiply A by k to get the coefficient of the derivative.

Step 3: Evaluate at $x = 0$.

$$8e^0 = 8$$

$$e^0 = 1.$$

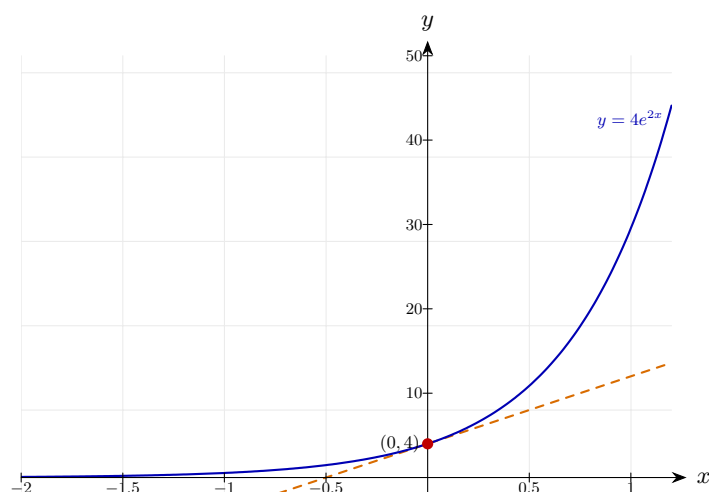
Step 4: Evaluate at $x = 1$.

$$8e^{2 \times 1} = 8e^2$$

Leave as an exact value.

Final answer: At $x = 0$: gradient = 8. At $x = 1$: gradient = $8e^2$.

Diagram.



The exponential function plotted with key points marked and the tangent (dashed) showing the gradient.

Exponentials 12 (e11-012)

Difficulty: Foundation / Marks: 4

Find the gradient of $f(x) = 7e^{5x}$ at the points $x = 0$ and $x = 2$. Leave your answers as exact values.

Worked solution.

Step 1: Identify $A = 7$ and $k = 5$.

$$f(x) = 7e^{5x}$$

Standard form Ae^{kx} .

Step 2: The gradient is kAe^{kx} .

$$f'(x) = 5 \times 7 \times e^{5x} = 35e^{5x}$$

Multiply the exponent by the coefficient.

Step 3: Gradient at $x = 0$.

$$f'(0) = 35e^0 = 35$$

$$e^0 = 1.$$

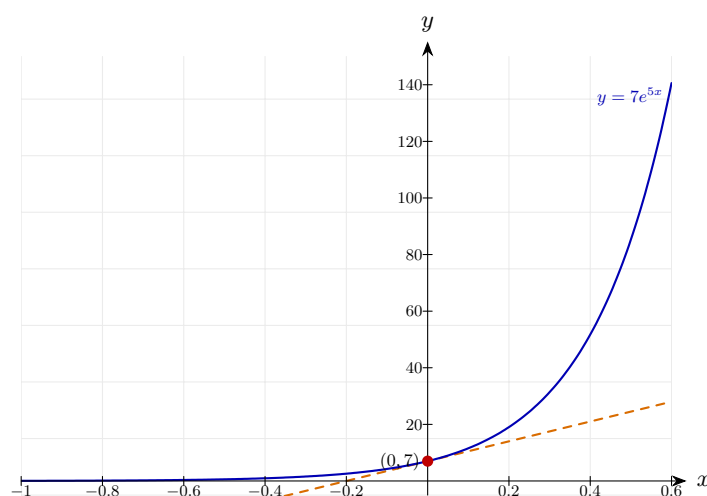
Step 4: Gradient at $x = 2$.

$$f'(2) = 35e^{10}$$

Exact value.

Final answer: At $x = 0$: gradient = 35. At $x = 2$: gradient = $35e^{10}$.

Diagram.



The exponential function plotted with key points marked and the tangent (dashed) showing the gradient.

Exponentials 13 (e11-013)

Difficulty: Foundation / Marks: 4

Find the gradient of $y = 3e^{-2x}$ at the points $x = 1$ and $x = -1$. Leave your answers as exact values.

Worked solution.

Step 1: Here $A = 3$ and $k = -2$.

$$y = 3e^{-2x}$$

The exponent is negative, so the function is decreasing.

Step 2: Compute the gradient formula.

$$\frac{dy}{dx} = (-2)(3)e^{-2x} = -6e^{-2x}$$

Multiply k by A .

Step 3: Gradient at $x = 1$.

$$-6e^{-2} = -\frac{6}{e^2}$$

The gradient is negative (decreasing function).

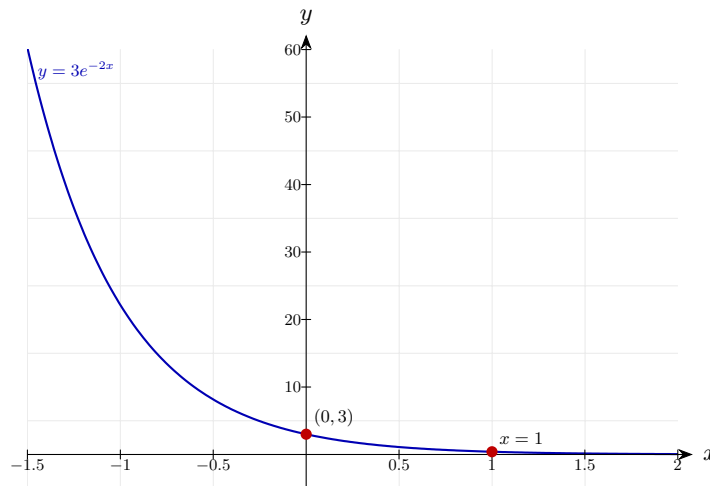
Step 4: Gradient at $x = -1$.

$$-6e^{-2(-1)} = -6e^2$$

Substituting $x = -1$.

Final answer: At $x = 1$: gradient $= -\frac{6}{e^2}$. At $x = -1$: gradient $= -6e^2$.

Diagram.



The exponential function plotted with key points marked.

Exponentials 14 (e11-014)

Difficulty: Foundation / Marks: 4

Find the gradient of $y = 6e^{4x}$ at the points $x = -1$ and $x = 0.5$. Leave your answers as exact values.

Worked solution.

Step 1: Identify $A = 6$, $k = 4$.

$$y = 6e^{4x}$$

Standard form.

Step 2: Gradient formula.

$$\frac{dy}{dx} = 4 \times 6 \times e^{4x} = 24e^{4x}$$

Multiply the coefficient by the exponent.

Step 3: Gradient at $x = -1$.

$$24e^{-4} = \frac{24}{e^4}$$

Substitute $x = -1$.

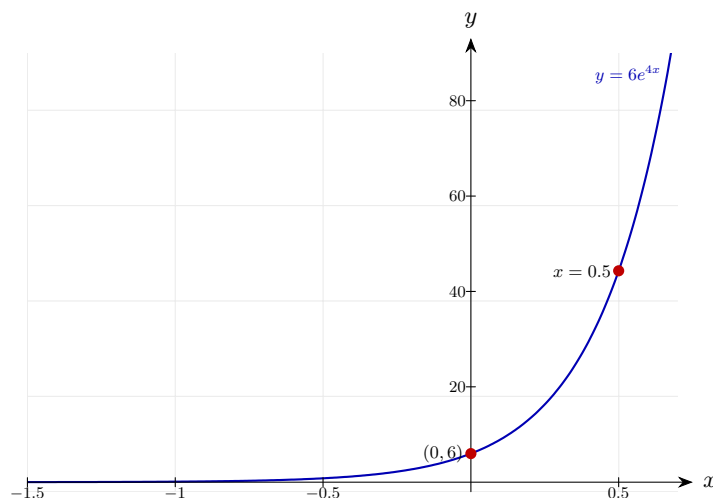
Step 4: Gradient at $x = 0.5$.

$$24e^{4 \times 0.5} = 24e^2$$

Substitute $x = 0.5$.

Final answer: At $x = -1$: gradient $= \frac{24}{e^4}$. At $x = 0.5$: gradient $= 24e^2$.

Diagram.



The exponential function plotted with key points marked.

Exponentials 15 (e11-015)

Difficulty: Foundation / Marks: 4

Find the gradient of $g(t) = 2.5e^{3t}$ at the points $t = 0$ and $t = 4$. Leave your answers as exact values.

Worked solution.

Step 1: Identify $A = 2.5$ and $k = 3$.

$$g(t) = 2.5e^{3t}$$

The variable is t instead of x — the same rule applies.

Step 2: Gradient formula.

$$g'(t) = 3 \times 2.5 \times e^{3t} = 7.5e^{3t}$$

$$kA = 3 \times 2.5 = 7.5.$$

Step 3: Gradient at $t = 0$.

$$g'(0) = 7.5e^0 = 7.5$$

$$e^0 = 1.$$

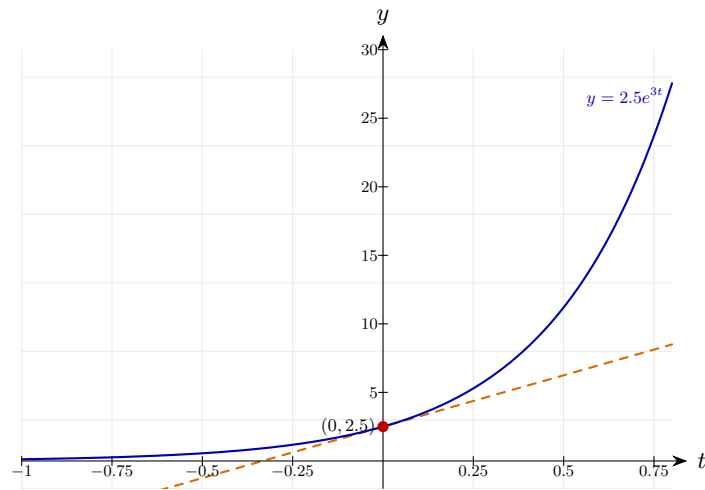
Step 4: Gradient at $t = 4$.

$$g'(4) = 7.5e^{12}$$

Exact value.

Final answer: At $t = 0$: gradient $= 7.5$. At $t = 4$: gradient $= 7.5e^{12}$.

Diagram.



The exponential function plotted with key points marked and the tangent (dashed) showing the gradient.

Exponentials 16 (e11-016)

Difficulty: Foundation / Marks: 3

Three graphs are shown. Match each of the following functions to the correct graph: $y = e^x$, $y = 2e^x$, $y = e^{2x}$. Justify your answers.

Worked solution.

Step 1: All three pass through or near $(0, 1)$ — but $y = 2e^x$ passes through $(0, 2)$.

$$y = 2e^0 = 2; \quad y = e^0 = 1; \quad y = e^0 = 1$$

The graph with y -intercept 2 is $y = 2e^x$.

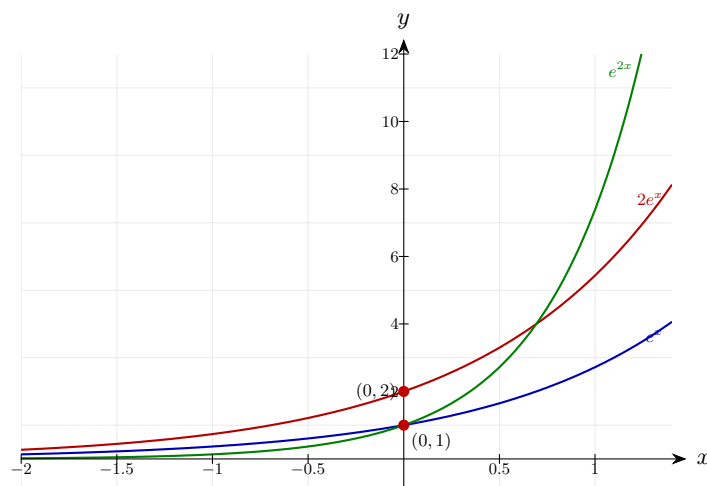
Step 2: Between $y = e^x$ and $y = e^{2x}$: both pass through $(0, 1)$, but $y = e^{2x}$ grows much faster.

$$e^{2x} = (e^x)^2 \text{ so it grows faster for } x > 0$$

The steeper graph is $y = e^{2x}$; the shallower one is $y = e^x$.

Final answer: Graph with y -intercept 2 $\rightarrow y = 2e^x$. Steepest remaining graph $\rightarrow y = e^{2x}$.
Remaining $\rightarrow y = e^x$.

Diagram.



The exponential function plotted with key points marked.

Exponentials 17 (e11-017)

Difficulty: Foundation / Marks: 3

Describe the transformation that maps $y = e^x$ onto $y = e^x + 3$. State the y -intercept and asymptote of $y = e^x + 3$.

Worked solution.

Step 1: Adding 3 outside the function shifts the graph up by 3 units.

$$y = e^x \rightarrow y = e^x + 3 \quad (\text{translation } +3 \text{ in } y\text{-direction})$$

This is a vertical translation.

Step 2: Find the y -intercept.

$$x = 0 : \quad y = e^0 + 3 = 1 + 3 = 4$$

y -intercept is $(0, 4)$.

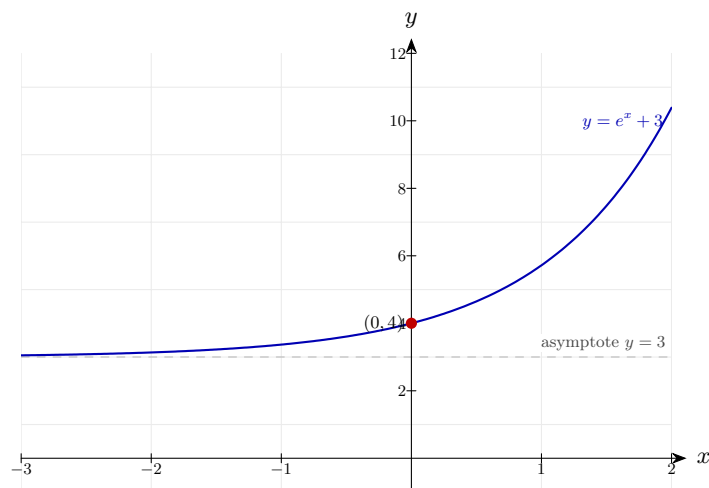
Step 3: The asymptote shifts up too.

$$\text{As } x \rightarrow -\infty, \quad e^x \rightarrow 0 \Rightarrow y \rightarrow 3$$

The original asymptote $y = 0$ shifts to $y = 3$.

Final answer: Vertical translation +3. y -intercept $(0, 4)$; asymptote $y = 3$.

Diagram.



The exponential function plotted with key points marked.

Exponentials 18 (e11-018)

Difficulty: Foundation / Marks: 3

Describe the transformation that maps $y = e^x$ onto $y = e^{x-2}$. State the y -intercept and the asymptote of $y = e^{x-2}$.

Worked solution.

Step 1: Replacing x with $x - 2$ shifts the graph 2 units to the right.

$$y = e^x \rightarrow y = e^{x-2} \quad (\text{translation } +2 \text{ in } x\text{-direction})$$

This is a horizontal translation of $+2$.

Step 2: Find the y -intercept.

$$x = 0 : \quad y = e^{0-2} = e^{-2} = \frac{1}{e^2}$$

The y -intercept is at $(0, \frac{1}{e^2})$.

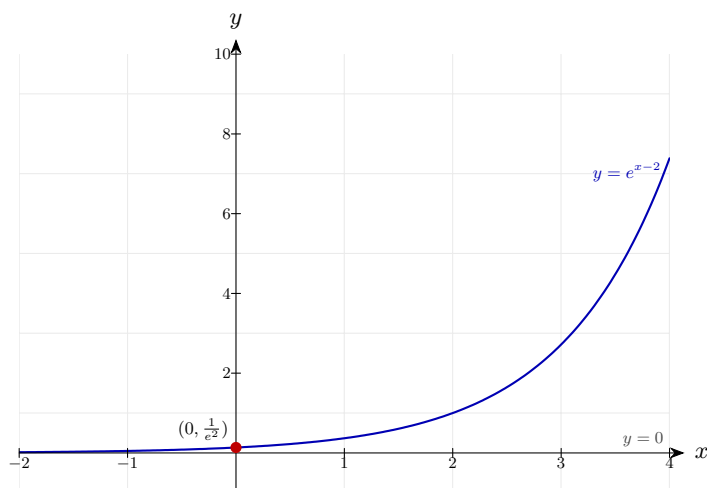
Step 3: The asymptote remains $y = 0$ (horizontal translation does not move the asymptote).

$$\text{As } x \rightarrow -\infty, \quad y \rightarrow 0$$

The asymptote is unchanged.

Final answer: Horizontal translation $+2$. y -intercept $(0, \frac{1}{e^2})$; asymptote $y = 0$.

Diagram.



The exponential function plotted with key points marked.

Exponentials 19 (e11-019)

Difficulty: Foundation / Marks: 4

For the function $y = 3e^x$:

- (a) State the y -intercept.
- (b) Describe the transformation from $y = e^x$.
- (c) Find the gradient of $y = 3e^x$ at $x = 0$.

Worked solution.

Step 1: Part (a): Set $x = 0$.

$$y = 3e^0 = 3 \Rightarrow (0, 3)$$

y -intercept is $(0, 3)$.

Step 2: Part (b): Multiplying by 3 stretches vertically by scale factor 3.

$$y = e^x \rightarrow y = 3e^x \quad (\text{vertical stretch, scale factor 3})$$

Every y -value is multiplied by 3.

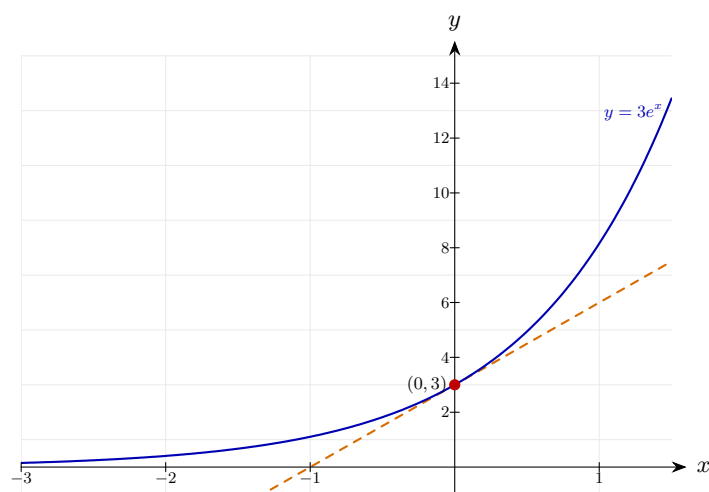
Step 3: Part (c): Gradient of $y = 3e^x$ is $3e^x$ (since $k = 1$).

$$\left. \frac{dy}{dx} \right|_{x=0} = 3e^0 = 3$$

The gradient equals the function value everywhere for $y = 3e^x$.

Final answer: (a) $(0, 3)$. (b) Vertical stretch by scale factor 3. (c) Gradient = 3.

Diagram.



The exponential function plotted with key points marked and the tangent (dashed) showing the gradient.

Exponentials 20 (e11-020)

Difficulty: Foundation / Marks: 5

On the same axes, sketch $y = e^x$, $y = e^x - 1$, and $y = -e^x$. Label all y -intercepts and asymptotes.

Worked solution.

Step 1: $y = e^x$: passes through $(0, 1)$, increasing, asymptote $y = 0$.

y -intercept: $(0, 1)$; asymptote: $y = 0$

Standard exponential.

Step 2: $y = e^x - 1$: same shape, shifted down 1 unit. Passes through $(0, 0)$; asymptote $y = -1$.

y -intercept: $(0, 0)$; asymptote: $y = -1$

Vertical translation of -1 . Note it crosses the origin!

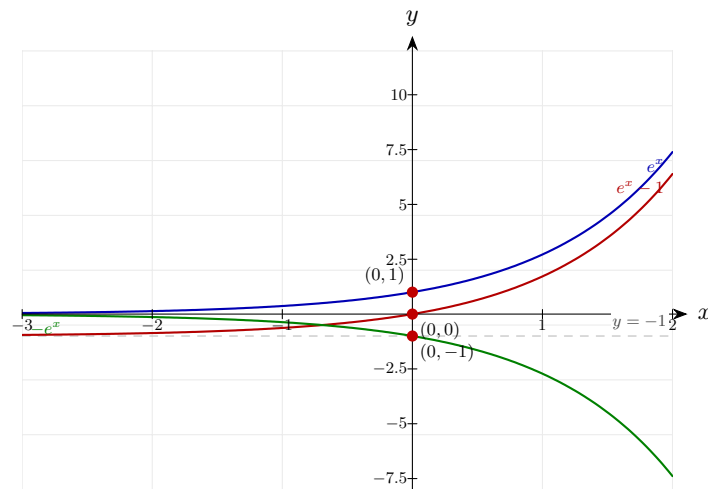
Step 3: $y = -e^x$: reflection of $y = e^x$ in the x -axis. Passes through $(0, -1)$; decreasing; asymptote $y = 0$ (approached from below).

y -intercept: $(0, -1)$; asymptote: $y = 0$

The curve lies entirely below the x -axis.

Final answer: $y = e^x$: $(0, 1)$, asymptote $y = 0$. $y = e^x - 1$: $(0, 0)$, asymptote $y = -1$.
 $y = -e^x$: $(0, -1)$, asymptote $y = 0$.

Diagram.



The exponential function plotted with key points marked.

Exponentials 21 (e11-021)

Difficulty: Foundation / Marks: 3

The graph of $y = Ae^{kx}$ has gradient $6e^{3x}$. Find the values of A and k .

Worked solution.

Step 1: The gradient of $y = Ae^{kx}$ is kAe^{kx} .

$$\frac{dy}{dx} = kAe^{kx}$$

Standard differentiation rule for this form.

Step 2: Match with the given gradient $6e^{3x}$.

$$kA = 6 \quad \text{and} \quad k = 3$$

Equate the exponents and the coefficients.

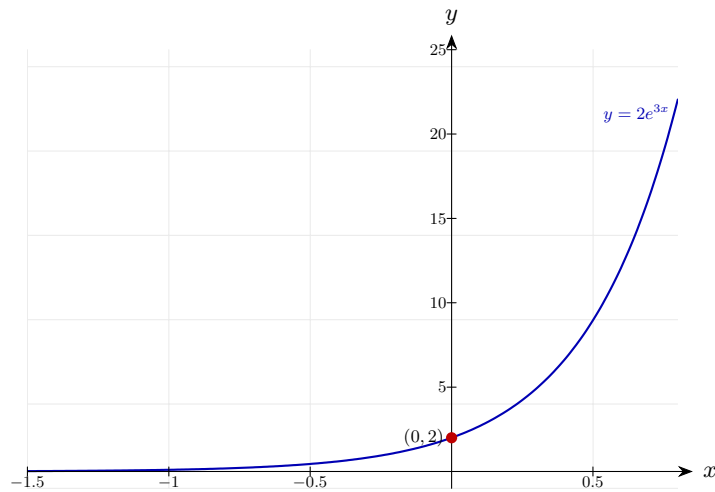
Step 3: Solve for A .

$$A = \frac{6}{k} = \frac{6}{3} = 2$$

$$kA = 6 \text{ and } k = 3 \Rightarrow A = 2.$$

Final answer: $k = 3$, $A = 2$, so $y = 2e^{3x}$.

Diagram.



The exponential function plotted with key points marked.

Exponentials 22 (e11-022)

Difficulty: Foundation / Marks: 5

The graph of $y = Ae^{kx}$ has gradient $20e^{4x}$. Find A and k , then find the exact y -coordinates when $x = 0$ and $x = 3$.

Worked solution.

Step 1: Gradient of $y = Ae^{kx}$ is kAe^{kx} .

$$kAe^{kx} = 20e^{4x}$$

Match the two expressions.

Step 2: Match the exponent: $k = 4$. Match the coefficient: $kA = 20 \Rightarrow A = 5$.

$$k = 4, \quad A = 5$$

So $y = 5e^{4x}$.

Step 3: y -coordinate at $x = 0$.

$$y = 5e^0 = 5$$

The y -intercept.

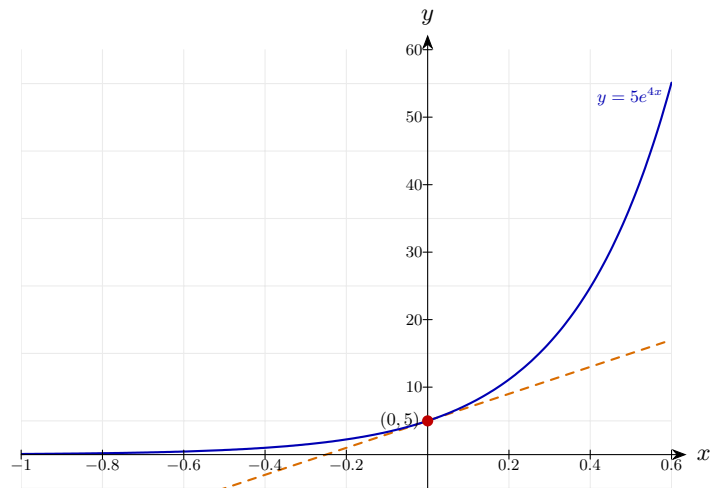
Step 4: y -coordinate at $x = 3$.

$$y = 5e^{12}$$

Exact value.

Final answer: $A = 5$, $k = 4$. At $x = 0$: $y = 5$. At $x = 3$: $y = 5e^{12}$.

Diagram.



The exponential function plotted with key points marked and the tangent (dashed) showing the gradient.

Exponentials 23 (e11-023)

Difficulty: Foundation / Marks: 4

The graph of $y = Ae^{kx}$ passes through $(0, 4)$ and has gradient 12 at $x = 0$. Find A and k .

Worked solution.

Step 1: Use the point $(0, 4)$ to find A .

$$Ae^{k \times 0} = Ae^0 = A = 4$$

At $x = 0$, $y = A$.

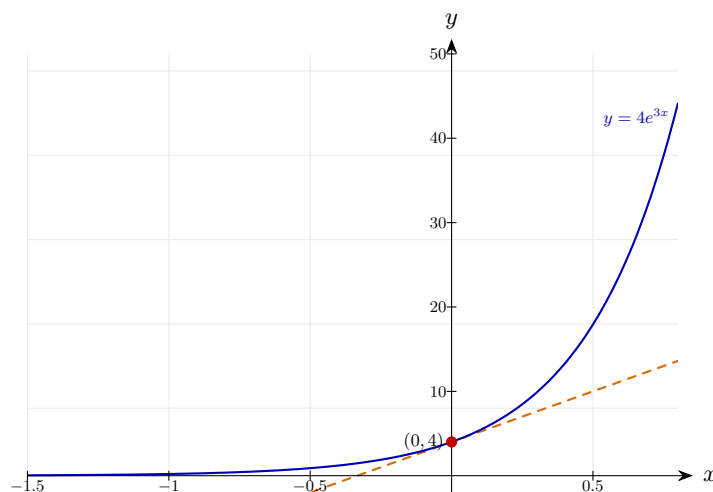
Step 2: Use the gradient at $x = 0$ to find k .

$$kAe^0 = kA = k \times 4 = 12 \Rightarrow k = 3$$

Gradient at $x = 0$ is kA .

Final answer: $A = 4$, $k = 3$, so $y = 4e^{3x}$.

Diagram.



The exponential function plotted with key points marked and the tangent (dashed) showing the gradient.

Exponentials 24 (e11-024)

Difficulty: Foundation / Marks: 4

The graph of $y = Ae^{kx}$ passes through $(0, 6)$ and the gradient at $x = 0$ is -18 . Find A and k , and state whether the function is increasing or decreasing.

Worked solution.

Step 1: From the y -intercept: $A = 6$.

$$y(0) = Ae^0 = A = 6$$

At $x = 0$, $e^0 = 1$.

Step 2: Gradient at $x = 0$ is $kA = -18$.

$$k \times 6 = -18 \Rightarrow k = -3$$

A negative k means the function is decreasing.

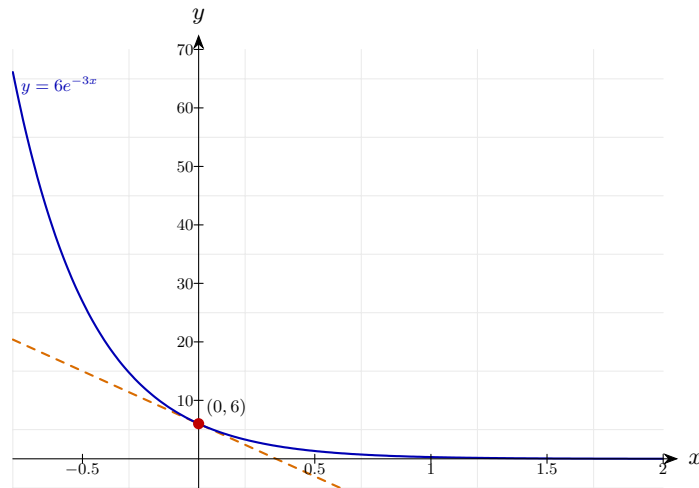
Step 3: State the nature.

$$k = -3 < 0 \Rightarrow \text{decreasing}$$

Since $k < 0$, the function decays exponentially.

Final answer: $A = 6$, $k = -3$, so $y = 6e^{-3x}$. The function is decreasing.

Diagram.



The exponential function plotted with key points marked and the tangent (dashed) showing the gradient.

Exponentials 25 (e11-025)

Difficulty: Foundation / Marks: 5

The graph of $y = Ae^{kx}$ has gradient $15e^{5x}$ at point $x = 2$. Find A and k , and hence find the exact y -coordinates at $x = 4$ and $x = 10$.

Worked solution.

Step 1: The gradient function is $kAe^{kx} = 15e^{5x}$.

$$k = 5, \quad kA = 15 \Rightarrow A = 3$$

Match coefficients and exponents.

Step 2: So $y = 3e^{5x}$.

$$y = 3e^{5x}$$

The equation of the exponential function.

Step 3: Find y at $x = 4$.

$$y = 3e^{20}$$

$$5 \times 4 = 20.$$

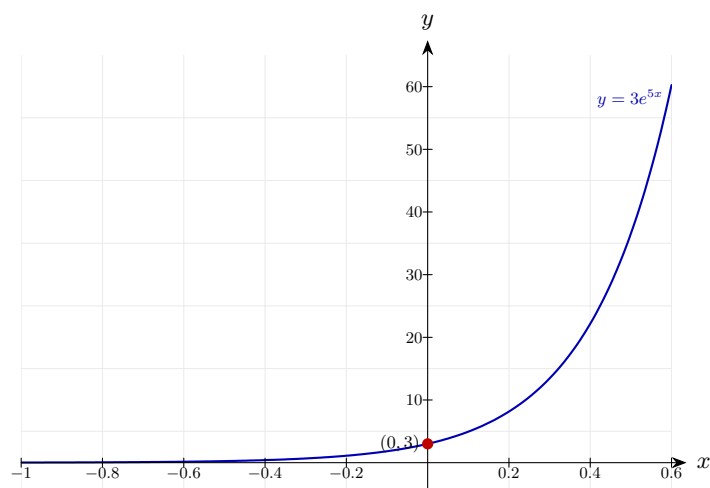
Step 4: Find y at $x = 10$.

$$y = 3e^{50}$$

$$5 \times 10 = 50.$$

Final answer: $A = 3, k = 5$. At $x = 4$: $y = 3e^{20}$. At $x = 10$: $y = 3e^{50}$.

Diagram.



The exponential function plotted with key points marked.

Exponentials 26 (e11-026)

Difficulty: Foundation / Marks: 4

Sketch the graph of $y = 2e^{-x}$, labelling the y -intercept and asymptote. Describe the two transformations from $y = e^x$.

Worked solution.

Step 1: Transformation 1: replace x with $-x$ to get $y = e^{-x}$.

$$y = e^x \rightarrow y = e^{-x} \quad (\text{reflection in } y\text{-axis})$$

Produces a decreasing exponential.

Step 2: Transformation 2: multiply by 2 to get $y = 2e^{-x}$.

$$y = e^{-x} \rightarrow y = 2e^{-x} \quad (\text{vertical stretch, scale factor 2})$$

Doubles all y -values.

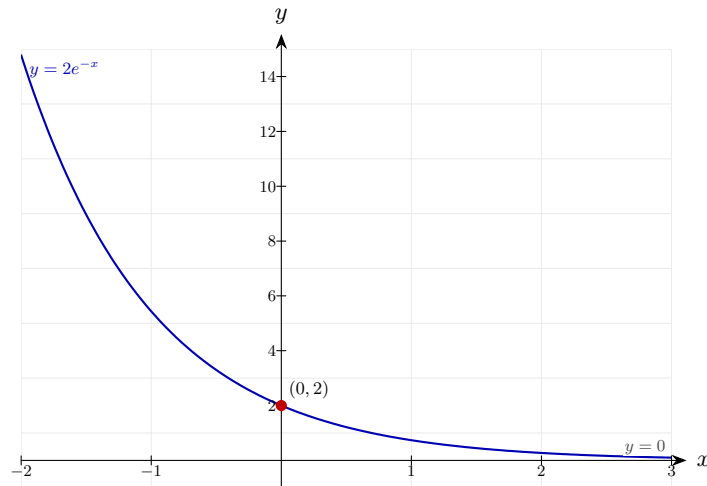
Step 3: Label the y -intercept and asymptote.

y -intercept: $(0, 2)$; asymptote: $y = 0$

$2e^0 = 2$. As $x \rightarrow \infty, y \rightarrow 0$.

Final answer: y -intercept $(0, 2)$; asymptote $y = 0$. Transformations: reflection in y -axis, then vertical stretch factor 2.

Diagram.



The exponential function plotted with key points marked.

Exponentials 27 (e11-027)

Difficulty: Foundation / Marks: 4

For the graph of $y = 5e^{2x} - 3$:

- (a) State the asymptote.
- (b) Find the y -intercept.
- (c) State the range of the function.

Worked solution.

Step 1: Part (a): As $x \rightarrow -\infty, 5e^{2x} \rightarrow 0$, so $y \rightarrow -3$.

Asymptote: $y = -3$

The -3 shifts the asymptote from $y = 0$ to $y = -3$.

Step 2: Part (b): Set $x = 0$.

$$y = 5e^0 - 3 = 5 - 3 = 2 \Rightarrow (0, 2)$$

$e^0 = 1$.

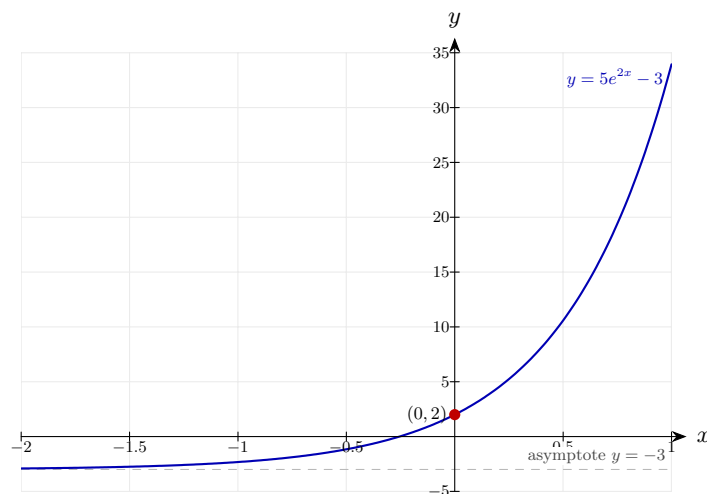
Step 3: Part (c): Since $5e^{2x} > 0$ always, we have $y > -3$ always.

$$y > -3 \quad (\text{i.e., range is } (-3, \infty))$$

The function never reaches its asymptote.

Final answer: (a) $y = -3$. (b) $(0, 2)$. (c) $y > -3$.

Diagram.



The exponential function plotted with key points marked.

Exponentials 28 (e11-028)

Difficulty: Foundation / Marks: 4

The function $f(x) = Ae^{kx}$ models the temperature ($^{\circ}\text{C}$) of a liquid cooling over time x (minutes). At $x = 0$ the temperature is 80°C , and the rate of cooling at $x = 0$ is $-8^{\circ}\text{C}/\text{min}$. Find A and k .

Worked solution.

Step 1: Use $f(0) = 80$ to find A .

$$f(0) = Ae^0 = A = 80$$

The initial temperature gives A .

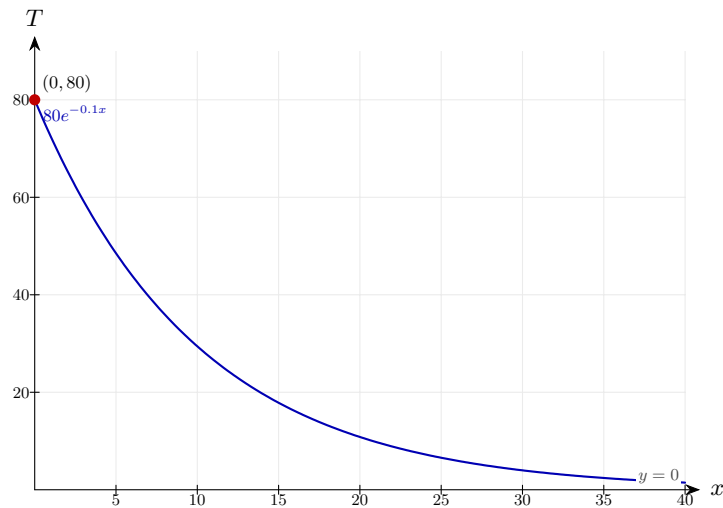
Step 2: The rate of cooling is the gradient. At $x = 0$, gradient $= kA = -8$.

$$k \times 80 = -8 \Rightarrow k = -\frac{1}{10} = -0.1$$

A negative gradient means the temperature is falling.

Final answer: $A = 80$, $k = -0.1$, so $f(x) = 80e^{-0.1x}$.

Diagram.



The exponential function plotted with key points marked.

Exponentials 29 (e11-029)

Difficulty: Foundation / Marks: 4

A population is modelled by $P = 500e^{0.04t}$, where t is time in years.

- State the initial population.
- Find the rate of growth of the population at $t = 0$.
- Is the population increasing or decreasing? Give a reason.

Worked solution.

Step 1: Part (a): Set $t = 0$.

$$P(0) = 500e^0 = 500$$

Initial population is 500.

Step 2: Part (b): Rate of growth = gradient = kAe^{kt} . At $t = 0$:

$$0.04 \times 500 \times e^0 = 20 \text{ individuals/year}$$

The population grows at 20 per year initially.

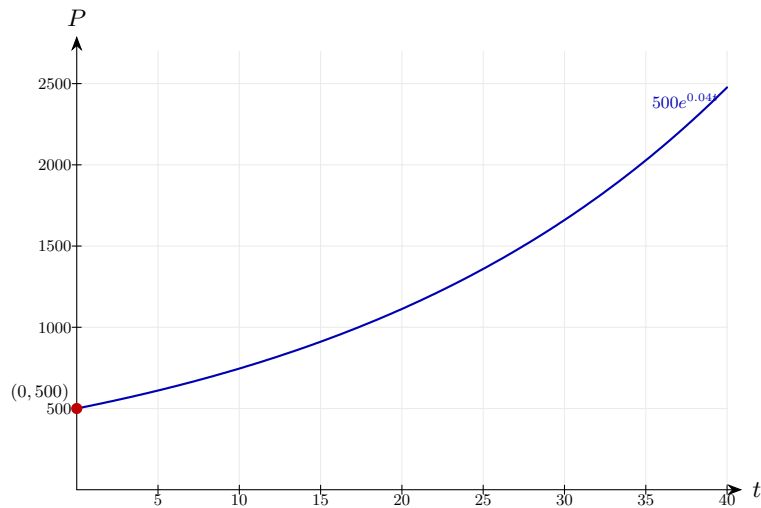
Step 3: Part (c): Since $k = 0.04 > 0$, the population is increasing.

$$k > 0 \Rightarrow P \text{ increases as } t \text{ increases}$$

A positive exponent means exponential growth.

Final answer: (a) 500. (b) 20 individuals/year. (c) Increasing, because $k = 0.04 > 0$.

Diagram.



The exponential function plotted with key points marked.

Exponentials 30 (e11-030)

Difficulty: Foundation / Marks: 4

Find the gradient of $y = 8e^{-3x}$ at the points $x = -2$ and $x = 0.5$. Leave your answers as exact values.

Worked solution.

Step 1: Identify $A = 8$ and $k = -3$.

$$y = 8e^{-3x}$$

Decreasing exponential.

Step 2: Gradient formula.

$$\frac{dy}{dx} = (-3)(8)e^{-3x} = -24e^{-3x}$$

$$kA = -24.$$

Step 3: Gradient at $x = -2$.

$$-24e^{-3(-2)} = -24e^6$$

Substitute $x = -2$.

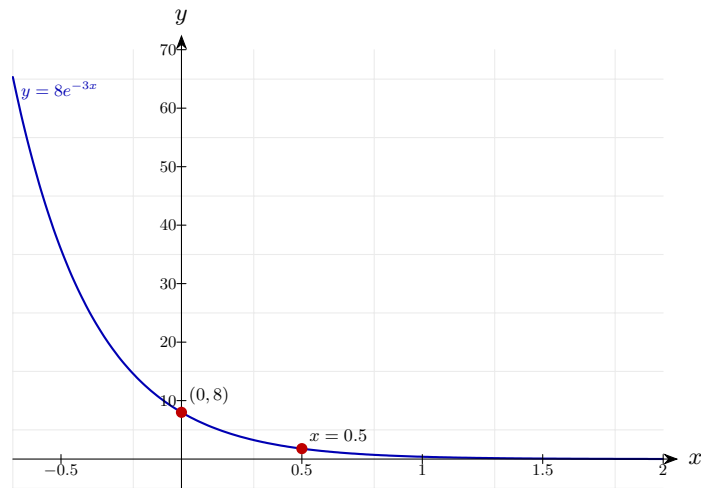
Step 4: Gradient at $x = 0.5$.

$$-24e^{-3(0.5)} = -24e^{-1.5} = -\frac{24}{e^{1.5}}$$

Substitute $x = 0.5$.

Final answer: At $x = -2$: gradient $= -24e^6$. At $x = 0.5$: gradient $= -\frac{24}{e^{3/2}}$.

Diagram.



The exponential function plotted with key points marked.

Exponentials 31 (e11-031)

Difficulty: Foundation / Marks: 6

The graph of $y = Ae^{kx}$ passes through the points $(0, 3)$ and has gradient 9 at $x = 0$.

- Find the values of A and k .
- Write down the gradient function.
- Find the exact gradient at $x = 2$.

Worked solution.

Step 1: Part (a): From the y -intercept $(0, 3)$.

$$A = 3$$

$$y(0) = A = 3.$$

Step 2: From gradient = 9 at $x = 0$: $kA = 9$.

$$k \times 3 = 9 \Rightarrow k = 3$$

So $y = 3e^{3x}$.

Step 3: Part (b): Gradient function.

$$\frac{dy}{dx} = 9e^{3x}$$

$kA = 9$, multiplied by e^{3x} .

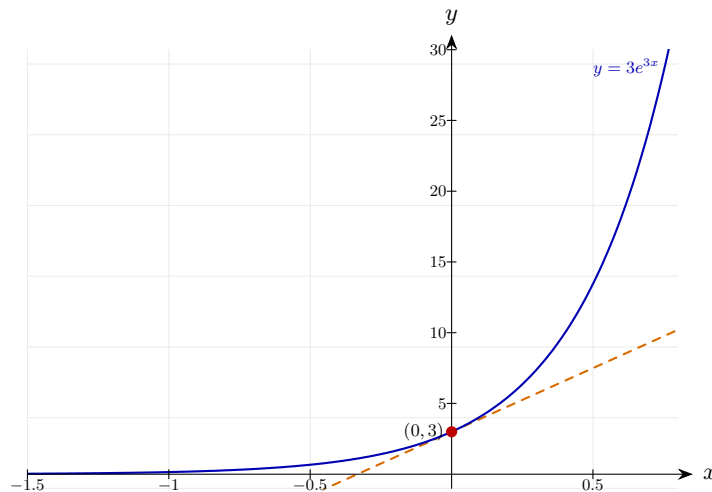
Step 4: Part (c): Gradient at $x = 2$.

$$9e^6$$

$$3 \times 2 = 6.$$

Final answer: (a) $A = 3$, $k = 3$. (b) $\frac{dy}{dx} = 9e^{3x}$. (c) Gradient = $9e^6$.

Diagram.



The exponential function plotted with key points marked and the tangent (dashed) showing the gradient.

Exponentials 32 (e11-032)

Difficulty: Foundation / Marks: 6

(a) On the same axes, sketch the graphs of $y = e^x$ and $y = 4 - e^x$ for $-2 \leq x \leq 2$. Label the intercepts of each graph.

(b) Find the coordinates of the point where the two graphs intersect.

Worked solution.

Step 1: Part (a): Key features of $y = e^x$.

$$y\text{-intercept: } (0, 1); \quad \text{asymptote: } y = 0$$

Standard increasing exponential.

Step 2: Key features of $y = 4 - e^x$.

$$y\text{-intercept: } (0, 4 - 1) = (0, 3); \quad x\text{-intercept: } e^x = 4 \Rightarrow x = \ln 4$$

$$y = 0 \Rightarrow e^x = 4 \Rightarrow x = \ln 4 \approx 1.39. \quad \text{Asymptote as } x \rightarrow -\infty: y \rightarrow 4.$$

Step 3: Part (b): Set the equations equal.

$$e^x = 4 - e^x \Rightarrow 2e^x = 4 \Rightarrow e^x = 2 \Rightarrow x = \ln 2$$

Solve for x by collecting e^x terms.

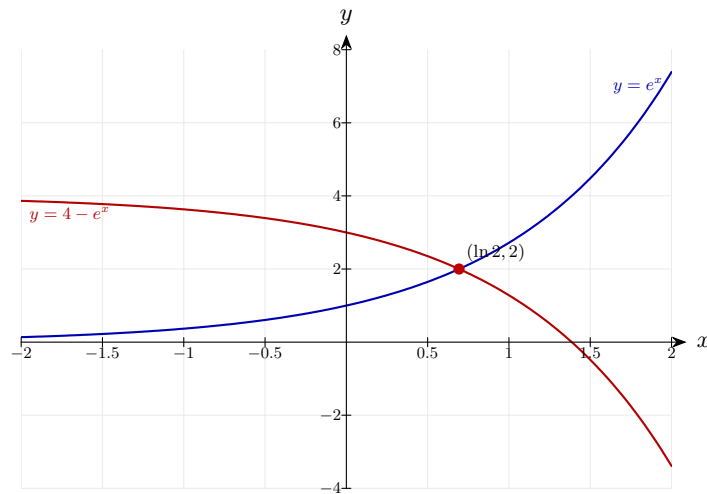
Step 4: Find y at the intersection.

$$y = e^{\ln 2} = 2 \Rightarrow \text{intersection at } (\ln 2, 2)$$

$$e^{\ln 2} = 2.$$

Final answer: (b) Intersection at $(\ln 2, 2)$.

Diagram.



The exponential function plotted with key points marked.

Exponentials 33 (e11-033)

Difficulty: Foundation / Marks: 5

The mass M grams of a radioactive substance at time t hours is modelled by $M = 120e^{-0.05t}$.

- Find the initial mass.
- Find the rate of decay at $t = 0$.
- Find the exact mass when $t = 20$.

Worked solution.

Step 1: Part (a): Set $t = 0$.

$$M(0) = 120e^0 = 120 \text{ g}$$

The initial mass is 120 g.

Step 2: Part (b): Rate of decay = gradient. $\frac{dM}{dt} = kAe^{kt}$. At $t = 0$:

$$kA = (-0.05)(120) = -6 \text{ g/hour}$$

Negative because the mass is decreasing.

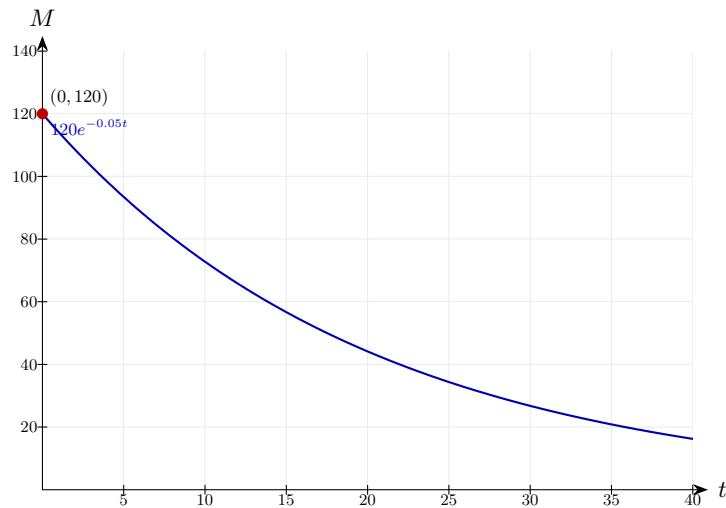
Step 3: Part (c): Substitute $t = 20$.

$$M(20) = 120e^{-0.05 \times 20} = 120e^{-1} = \frac{120}{e}$$

$$-0.05 \times 20 = -1.$$

Final answer: (a) 120 g. (b) -6 g/hour. (c) $M = \frac{120}{e}$ g.

Diagram.



The exponential function plotted with key points marked.

Exponentials 34 (e11-034)

Difficulty: Foundation / Marks: 6

The graph of $y = Ae^{kx}$ has gradient $12e^{2x}$ at the point $x = 2$. Find A and k , and hence find the exact y -coordinates when $x = 4$ and $x = 12$.

Worked solution.

Step 1: The gradient of $y = Ae^{kx}$ is kAe^{kx} . Match with $12e^{2x}$.

$$k = 2, \quad kA = 12 \Rightarrow A = 6$$

Equate exponents and coefficients.

Step 2: So $y = 6e^{2x}$.

$$y = 6e^{2x}$$

The equation of the curve.

Step 3: y -coordinate at $x = 4$.

$$y = 6e^8$$

$$2 \times 4 = 8.$$

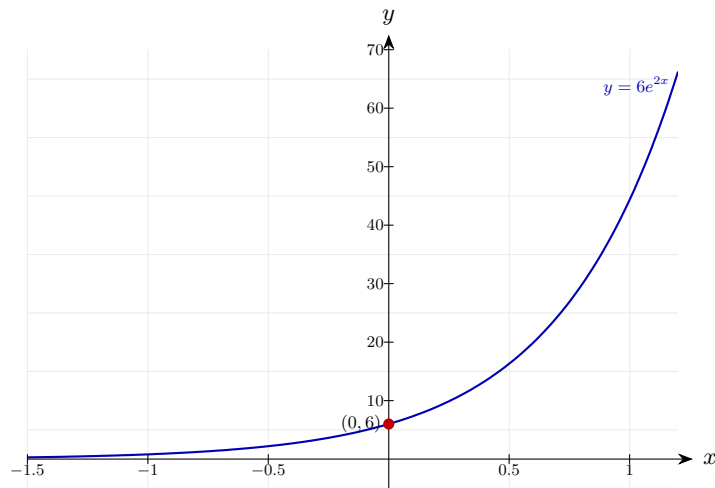
Step 4: y -coordinate at $x = 12$.

$$y = 6e^{24}$$

$$2 \times 12 = 24.$$

Final answer: $A = 6$, $k = 2$. At $x = 4$: $y = 6e^8$. At $x = 12$: $y = 6e^{24}$.

Diagram.



The exponential function plotted with key points marked.

Exponentials 35 (e11-035)

Difficulty: Foundation / Marks: 7

(a) Sketch the graph of $y = 3e^{2x} - 6$, labelling the y -intercept, any x -intercept, and the asymptote.

(b) State the range of the function.

(c) Find the gradient of the curve at $x = 0$.

Worked solution.

Step 1: Part (a): Find the asymptote.

$$\text{As } x \rightarrow -\infty, \quad 3e^{2x} \rightarrow 0 \Rightarrow y \rightarrow -6$$

Asymptote: $y = -6$.

Step 2: Find the y -intercept.

$$x = 0 : \quad y = 3e^0 - 6 = 3 - 6 = -3 \Rightarrow (0, -3)$$

$$e^0 = 1.$$

Step 3: Find the x -intercept (where $y = 0$).

$$3e^{2x} = 6 \Rightarrow e^{2x} = 2 \Rightarrow 2x = \ln 2 \Rightarrow x = \frac{\ln 2}{2}$$

Takes logarithms to solve for x .

Step 4: Sketch: increasing curve from asymptote $y = -6$, crossing x -axis at $x = \frac{\ln 2}{2} \approx 0.35$, y -intercept at -3 .

$$\text{x-intercept: } \left(\frac{\ln 2}{2}, 0\right)$$

Label all three key points/lines.

Step 5: Part (b): Since $3e^{2x} > 0$ always, $y > -6$ always.

$$\text{Range: } y > -6$$

The function never reaches its asymptote.

Step 6: Part (c): Gradient function.

$$\frac{dy}{dx} = 2 \times 3 \times e^{2x} = 6e^{2x}$$

$$kA = 2 \times 3 = 6.$$

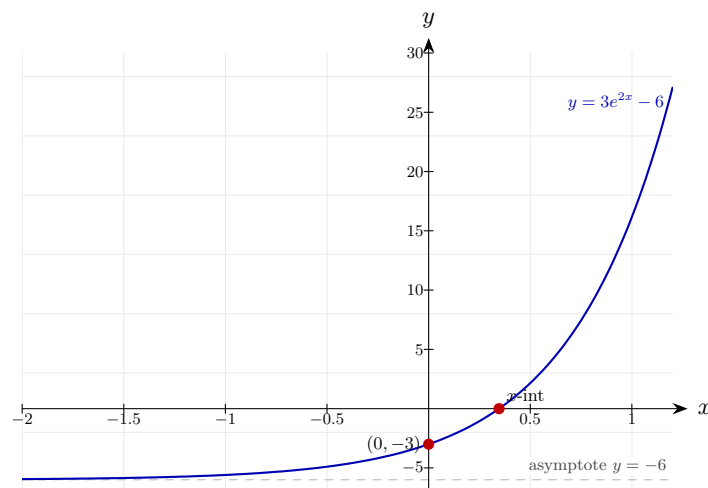
Step 7: Gradient at $x = 0$.

$$6e^0 = 6$$

The curve has gradient 6 at the y -intercept.

Final answer: (a) Asymptote $y = -6$, y -intercept $(0, -3)$, x -intercept $\left(\frac{\ln 2}{2}, 0\right)$. (b) $y > -6$.
(c) Gradient = 6.

Diagram.



The exponential function plotted with key points marked.

Exponentials 36 (e11-036)

Difficulty: Foundation / Marks: 1

Evaluate 2^5 .

Worked solution.

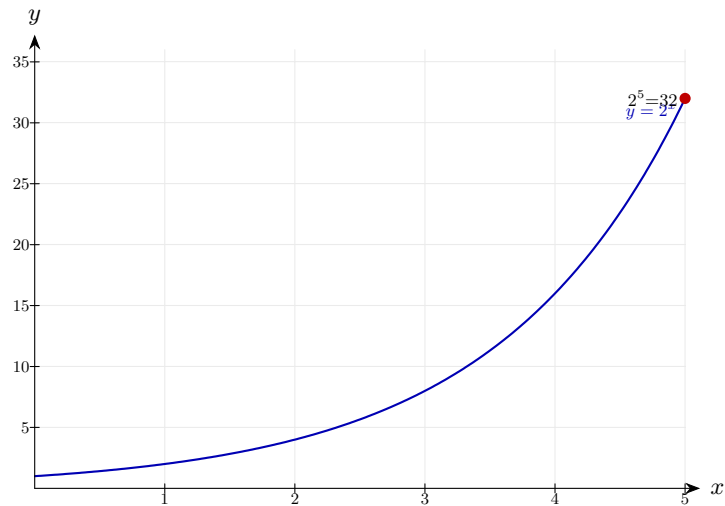
Calculate

$$2^5 = 32$$

2 multiplied by itself 5 times.

Final answer: 32

Diagram.



The exponential function plotted with key points marked.

Exponentials 37 (e11-037)

Difficulty: Foundation / Marks: 2

Simplify $3^2 \times 3^4$.

Worked solution.

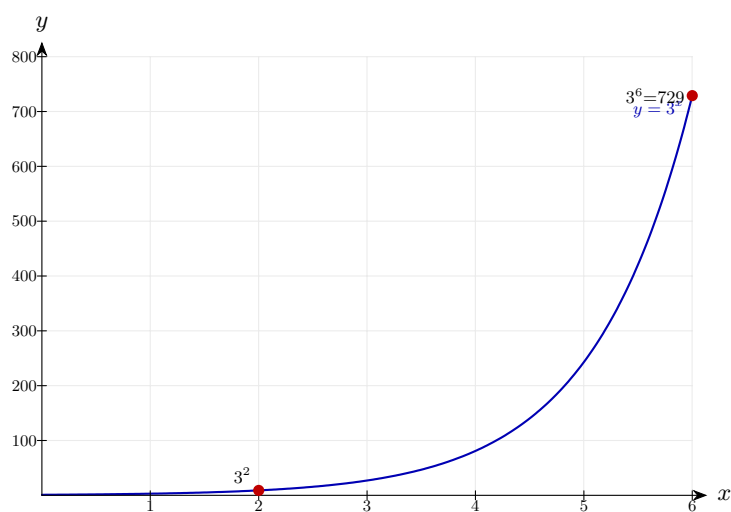
Add indices

$$3^{2+4} = 3^6 = 729$$

Same base: add powers.

Final answer: $3^6 = 729$

Diagram.



The exponential function plotted with key points marked.

Exponentials 38 (e11-038)

Difficulty: Foundation / Marks: 2

Simplify $\frac{5^7}{5^3}$.

Worked solution.

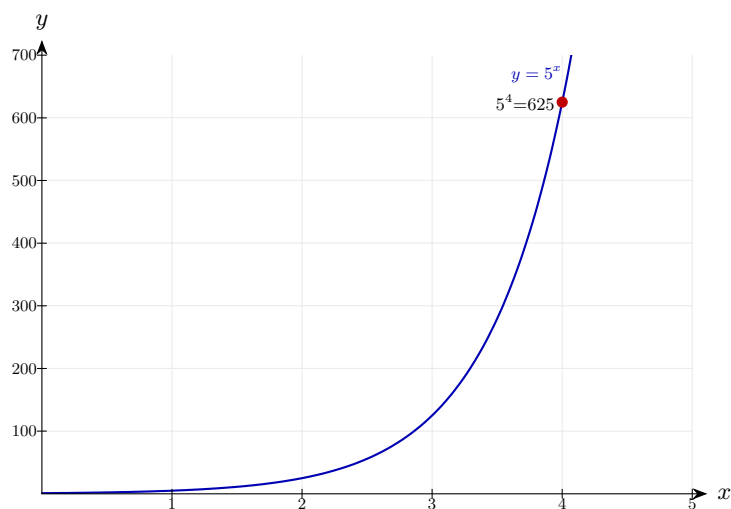
Subtract indices

$$5^{7-3} = 5^4 = 625$$

Dividing powers of the same base subtracts the exponents: $\frac{a^m}{a^n} = a^{m-n}$. Then $5^4 = 5 \times 5 \times 5 \times 5 = 625$.

Final answer: $5^4 = 625$

Diagram.



The exponential function plotted with key points marked.

Exponentials 39 (e11-039)

Difficulty: Foundation / Marks: 2

Evaluate 4^{-2} .

Worked solution.

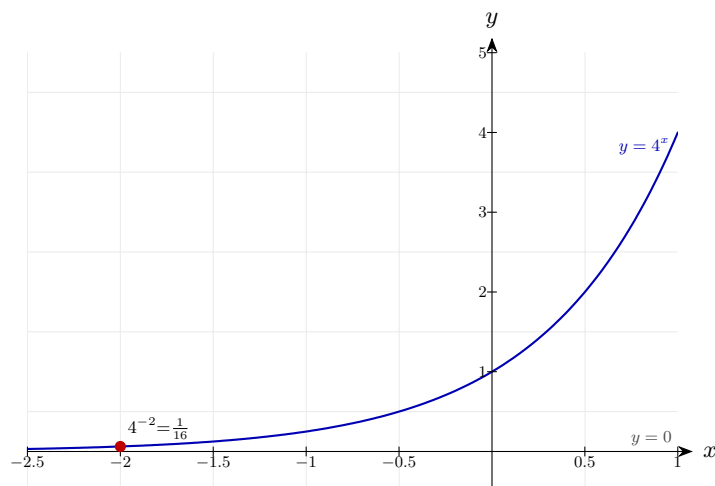
Negative index

$$4^{-2} = \frac{1}{4^2} = \frac{1}{16}$$

A negative index means take the reciprocal: $a^{-n} = \frac{1}{a^n}$. Pitfall: 4^{-2} is not -8 or -16 — the minus sign in the exponent never makes the result negative for a positive base.

Final answer: $\frac{1}{16}$

Diagram.



The exponential function plotted with key points marked.

Exponentials 40 (e11-040)

Difficulty: Foundation / Marks: 2

Evaluate $8^{\frac{2}{3}}$.

Worked solution.

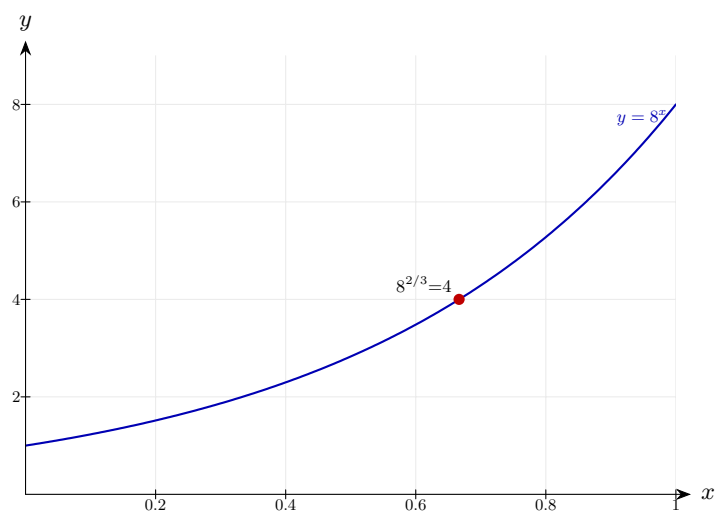
Fractional index

$$8^{2/3} = (8^{1/3})^2 = 2^2 = 4$$

Cube root of 8 is 2, then square.

Final answer: 4

Diagram.



The exponential function plotted with key points marked.

Exponentials 41 (e11-041)

Difficulty: Foundation / Marks: 3

Sketch $y = 2^x$ for $-3 \leq x \leq 3$. State the y-intercept and asymptote.

Worked solution.

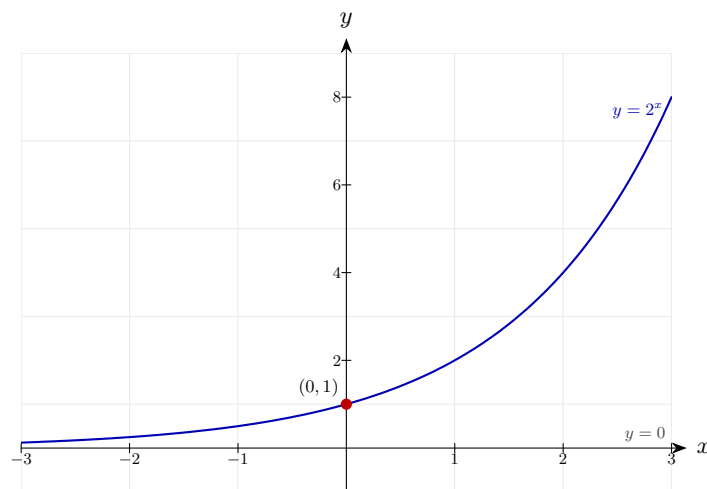
Key features

y-int: $(0, 1)$; asymptote: $y = 0$

Exponential growth; always positive.

Final answer: y-int $(0, 1)$; asymptote $y = 0$

Diagram.



The exponential function plotted with key points marked.

Exponentials 42 (e11-042)

Difficulty: Foundation / Marks: 3

Sketch $y = 3^{-x}$. Describe the transformation from $y = 3^x$.

Worked solution.

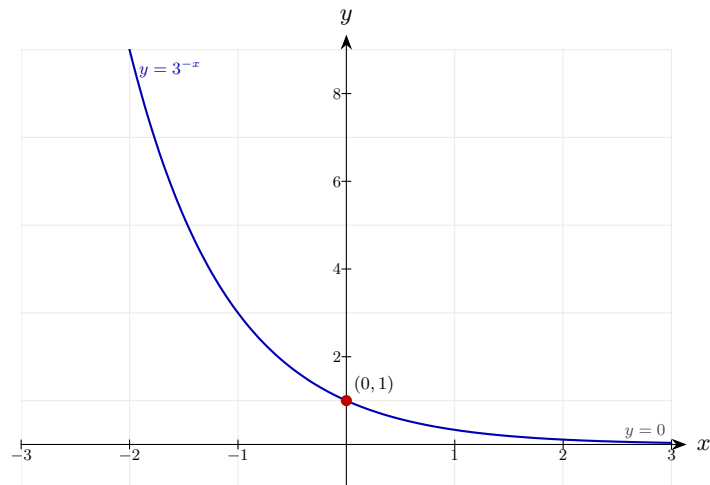
Reflection in y-axis

$$y = 3^{-x} = \left(\frac{1}{3}\right)^x$$

Exponential decay.

Final answer: Reflection in y-axis; exponential decay

Diagram.



The exponential function plotted with key points marked.

Exponentials 43 (e11-043)

Difficulty: Foundation / Marks: 2

Solve $2^x = 16$.

Worked solution.

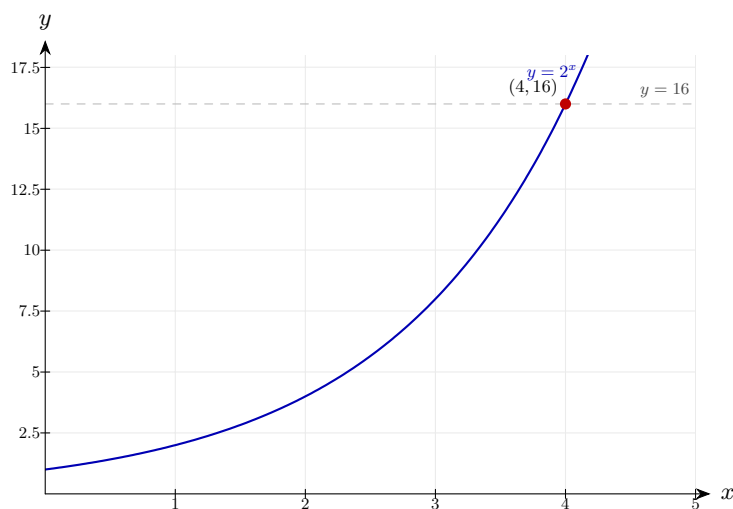
Write 16 as power of 2

$$2^x = 2^4 \implies x = 4$$

Rewrite the right-hand side using the same base as the left, then equate the exponents (because a^x is a one-to-one function for $a > 0$, $a \neq 1$). Since $16 = 2^4$, the equation forces $x = 4$.

Final answer: $x = 4$

Diagram.



The exponential function plotted with key points marked.

Exponentials 44 (e11-044)

Difficulty: Foundation / Marks: 2

Solve $3^x = \frac{1}{27}$.

Worked solution.

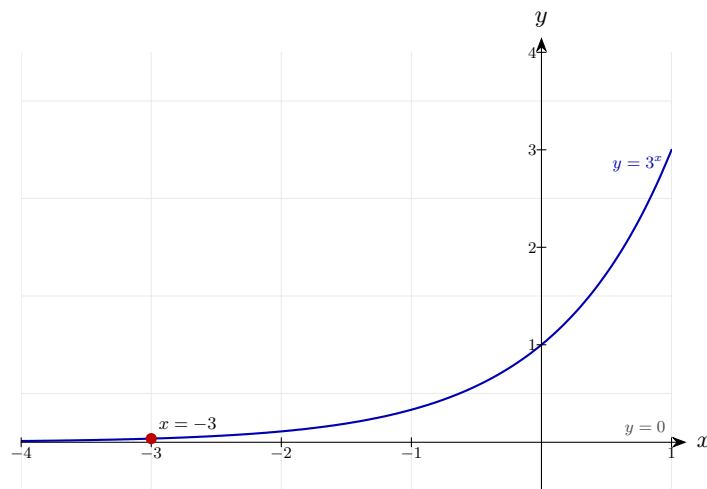
Write as power of 3

$$3^x = 3^{-3} \implies x = -3$$

$27 = 3^3$, so $1/27 = 3^{-3}$.

Final answer: $x = -3$

Diagram.



The exponential function plotted with key points marked.

Exponentials 45 (e11-045)

Difficulty: Foundation / Marks: 3

Solve $5^{2x} = 125$.

Worked solution.

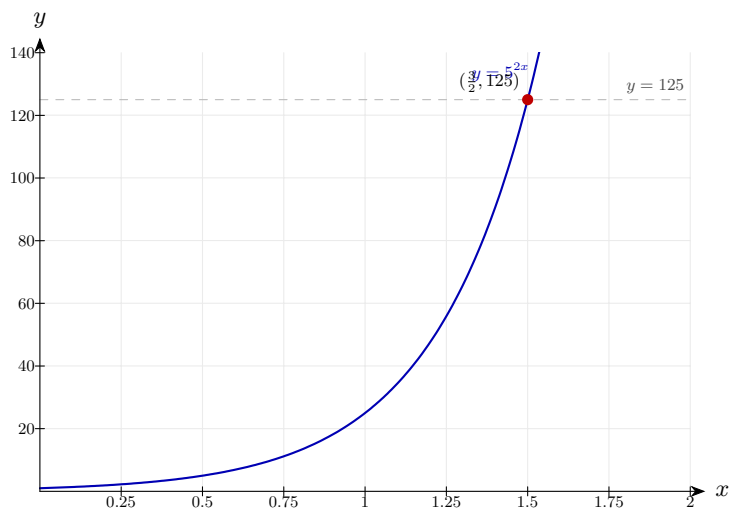
Write 125 as power of 5

$$5^{2x} = 5^3 \implies 2x = 3 \implies x = \frac{3}{2}$$

Rewrite both sides with the same base, then equate the exponents. Note $125 = 5^3$, and dividing both sides of $2x = 3$ by 2 gives the answer as a fraction — leave it as $\frac{3}{2}$ for an exact answer.

Final answer: $x = \frac{3}{2}$

Diagram.



The exponential function plotted with key points marked.

Exponentials 46 (e11-046)

Difficulty: Foundation / Marks: 2

The value of an investment is modelled by $V = 1000 \times 1.05^t$ where t is years. Find the value after 10 years.

Worked solution.

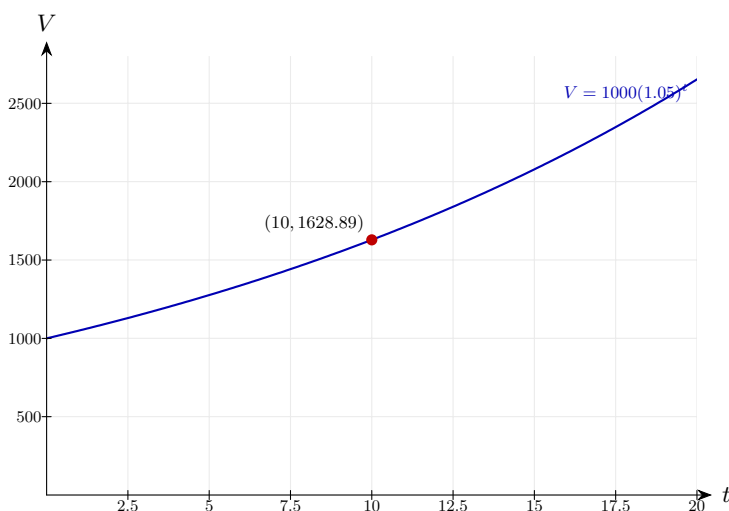
Substitute $t = 10$

$$V = 1000 \times 1.05^{10} = 1000 \times 1.6289 = 1628.89$$

For compound growth at 5% per year, each year multiplies the value by 1.05, so after 10 years the multiplier is $1.05^{10} \approx 1.6289$. Keep at least 4 d.p. on the intermediate multiplier before rounding the final answer to 2 d.p.

Final answer: 1628.89 pounds

Diagram.



The exponential function plotted with key points marked.

Exponentials 47 (e11-047)

Difficulty: Foundation / Marks: 1

Write e^3 to 3 significant figures.

Worked solution.

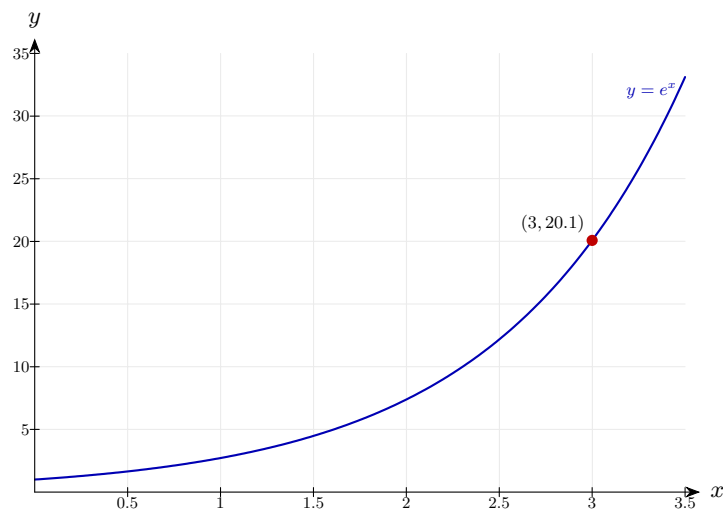
Calculator

$$e^3 = 20.1$$

Using the e^x button on a calculator: $e^3 \approx 20.0855$, which rounds to 20.1 (3 s.f.). Recall $e \approx 2.718$ is the natural base — the unique value for which the gradient of $y = e^x$ equals e^x itself.

Final answer: 20.1

Diagram.



The exponential function plotted with key points marked.

Exponentials 48 (e11-048)

Difficulty: Foundation / Marks: 3

Sketch $y = e^x$ and $y = e^{-x}$ on the same axes. State their point of intersection.

Worked solution.

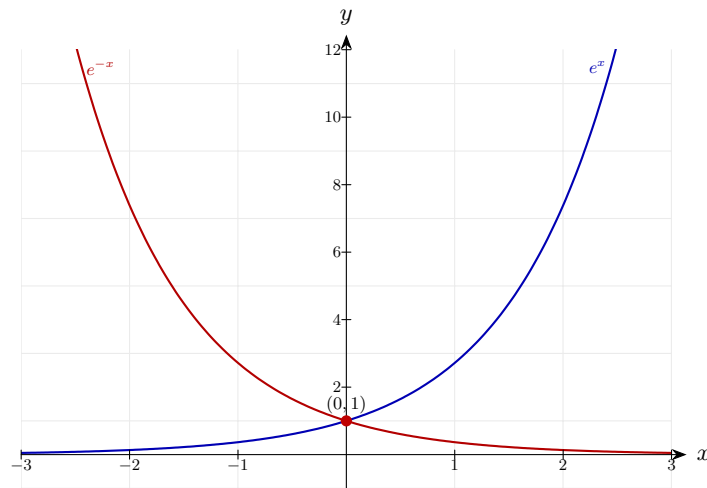
Intersection

$$e^x = e^{-x} \implies e^{2x} = 1 \implies x = 0, y = 1$$

Multiplying both sides by e^x (which is always positive) gives $e^{2x} = 1$, so $2x = 0$. Both curves pass through $(0, 1)$ because any base to the power 0 is 1 — and they are reflections of each other in the y -axis, so this is their only meeting point.

Final answer: Intersection at $(0, 1)$

Diagram.



The exponential function plotted with key points marked.

Exponentials 49 (e11-049)

Difficulty: Foundation / Marks: 1

Simplify $e^{2x} \times e^{3x}$.

Worked solution.

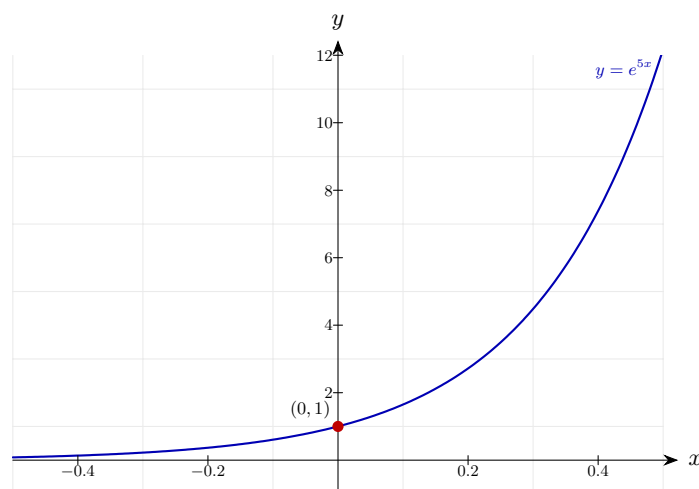
Add indices

$$e^{2x+3x} = e^{5x}$$

When multiplying powers of the same base, add the exponents: $a^m \times a^n = a^{m+n}$. The base e behaves like any other base under the laws of indices.

Final answer: e^{5x}

Diagram.



The exponential function plotted with key points marked.

Exponentials 50 (e11-050)

Difficulty: Foundation / Marks: 1

Simplify $\frac{e^{4x}}{e^x}$.

Worked solution.

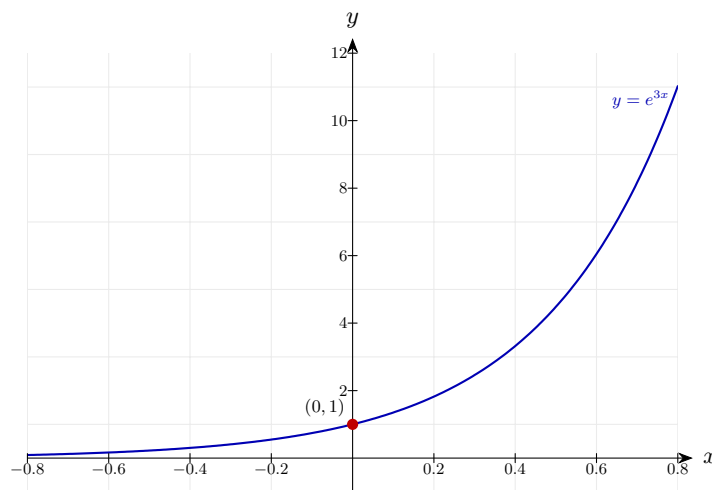
Subtract indices

$$e^{4x-x} = e^{3x}$$

Dividing powers of the same base subtracts the exponents: $\frac{a^m}{a^n} = a^{m-n}$. Here $4x - x = 3x$.

Final answer: e^{3x}

Diagram.



The exponential function plotted with key points marked.

Exponentials 51 (e11-051)

Difficulty: Foundation / Marks: 3

A population of bacteria doubles every 3 hours. Starting with 500 bacteria, find the population after 12 hours.

Worked solution.

Step 1: Number of doublings

$$12 \div 3 = 4 \text{ doublings}$$

Each doubling period is 3 hours, so in 12 hours the population doubles $\frac{12}{3} = 4$ times. This is the exponent in the growth factor.

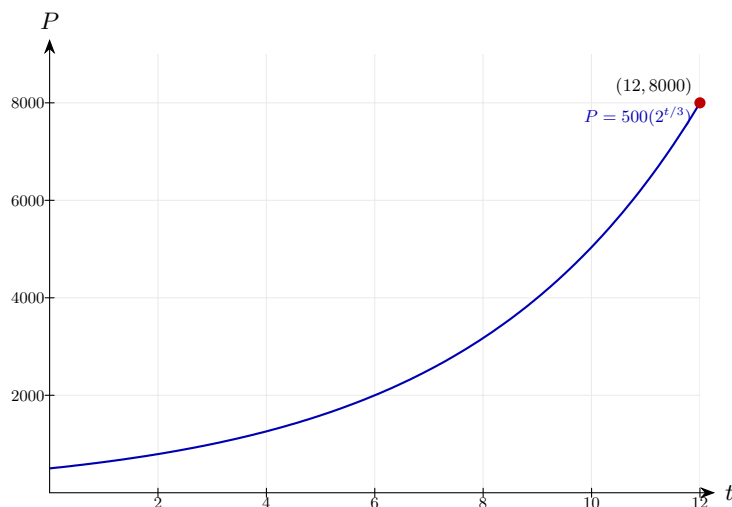
Step 2: Calculate

$$P = 500 \times 2^4 = 500 \times 16 = 8000$$

Multiply the starting amount by 2 for each doubling, i.e. by $2^4 = 16$. The underlying model is $P = 500 \times 2^{t/3}$ — the discrete-doubling viewpoint is just this evaluated at $t = 12$.

Final answer: 8000 bacteria

Diagram.



The exponential function plotted with key points marked.

Exponentials 52 (e11-052)

Difficulty: Foundation / Marks: 3

Solve $4^x = 8$. Give your answer as a fraction.

Worked solution.

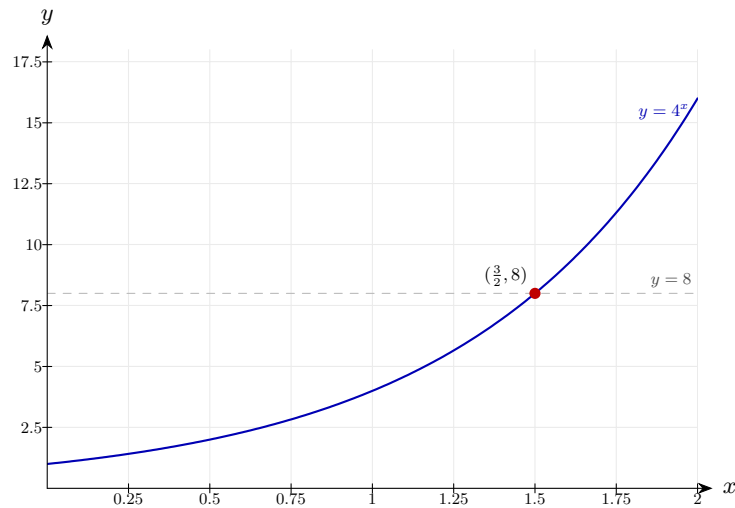
Write both as powers of 2

$$(2^2)^x = 2^3 \implies 2^{2x} = 2^3 \implies 2x = 3 \implies x = \frac{3}{2}$$

Rewrite both sides with the same base so the exponents can be equated. Using the power-of-a-power rule $(a^m)^n = a^{mn}$, $4^x = (2^2)^x = 2^{2x}$. Pitfall: do not write 4^x as $2^x \cdot 2$ — the entire base is squared, not multiplied.

Final answer: $x = \frac{3}{2}$

Diagram.



The exponential function plotted with key points marked.

Exponentials 53 (e11-053)

Difficulty: Foundation / Marks: 3

Solve $9^x = 27$.

Worked solution.

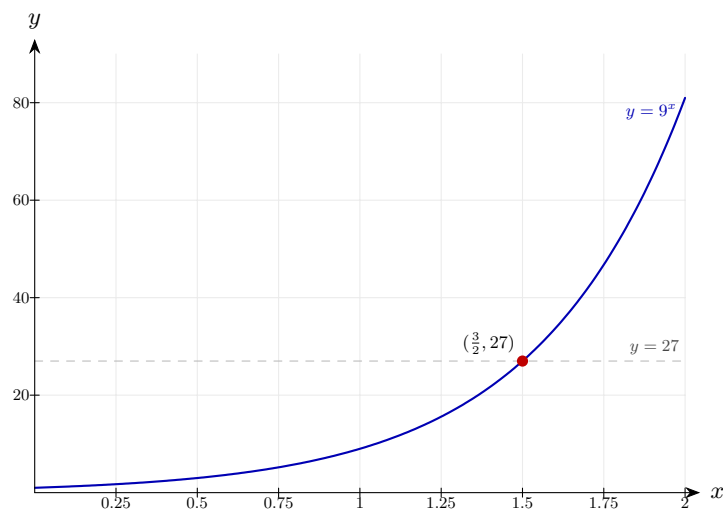
Write as powers of 3

$$(3^2)^x = 3^3 \implies 3^{2x} = 3^3 \implies x = \frac{3}{2}$$

Express both sides with base 3 using $9 = 3^2$ and $27 = 3^3$. Apply $(a^m)^n = a^{mn}$, then equate exponents:
 $2x = 3 \Rightarrow x = \frac{3}{2}$.

Final answer: $x = \frac{3}{2}$

Diagram.



The exponential function plotted with key points marked.

Exponentials 54 (e11-054)

Difficulty: Foundation / Marks: 2

Describe the transformation that maps $y = e^x$ to $y = e^x + 3$.

Worked solution.

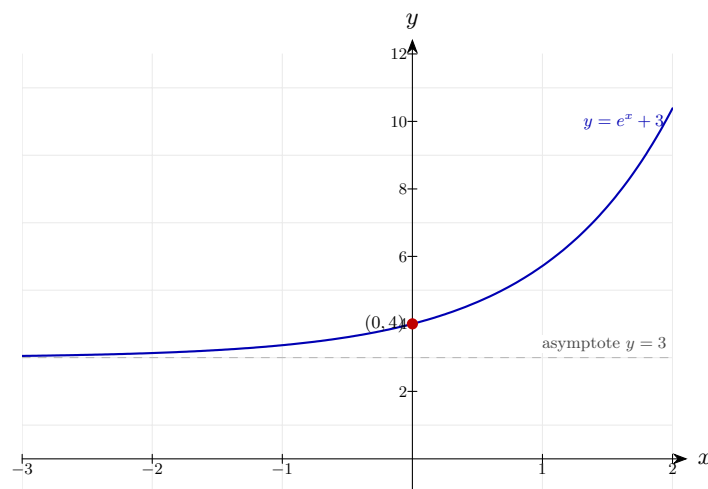
Translation

Translation 3 units up

Asymptote moves from $y = 0$ to $y = 3$.

Final answer: Translation 3 units up; new asymptote $y = 3$

Diagram.



The exponential function plotted with key points marked.

Exponentials 55 (e11-055)

Difficulty: Foundation / Marks: 2

Describe the transformation that maps $y = e^x$ to $y = e^{x-2}$.

Worked solution.

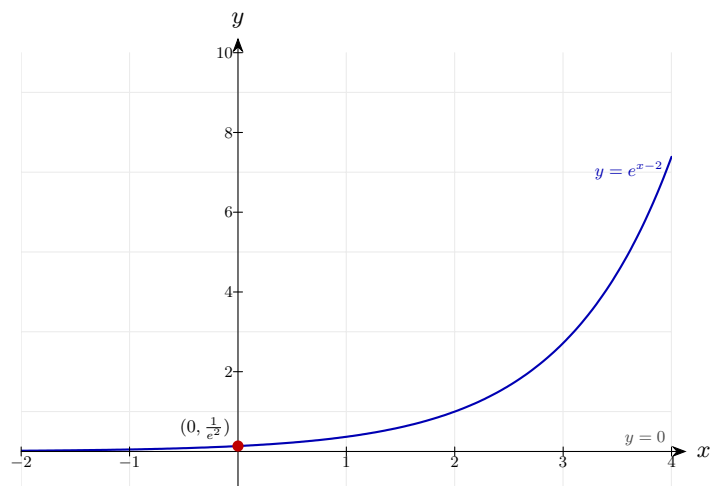
Translation

Translation 2 units right

Replacing x with $x - 2$ inside the function shifts the graph 2 units in the positive x -direction. The asymptote stays at $y = 0$ since horizontal translations do not move horizontal asymptotes. Pitfall: $x - 2$ gives a shift to the right, not the left.

Final answer: Translation 2 units right

Diagram.



The exponential function plotted with key points marked.

Exponentials 56 (e11-056)

Difficulty: Foundation / Marks: 2

Describe the transformation that maps $y = e^x$ to $y = 2e^x$.

Worked solution.

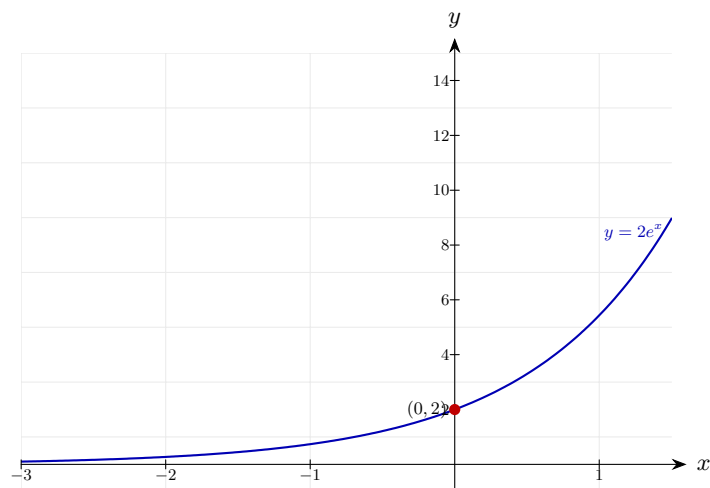
Stretch

Vertical stretch, scale factor 2

y-intercept moves from 1 to 2.

Final answer: Vertical stretch scale factor 2

Diagram.



The exponential function plotted with key points marked.

Exponentials 57 (e11-057)

Difficulty: Foundation / Marks: 2

A radioactive substance decays so that the mass m grams after t years is $m = 200e^{-0.1t}$. Find the mass after 5 years.

Worked solution.

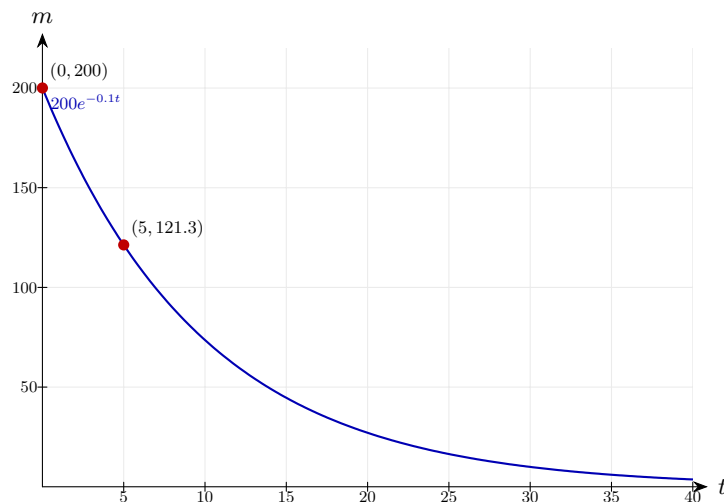
Substitute

$$m = 200e^{-0.5} = 200 \times 0.6065 = 121.3 \text{ g}$$

Put $t = 5$ into the model: the exponent becomes $-0.1 \times 5 = -0.5$, and $e^{-0.5} \approx 0.6065$. The base e appears because radioactive decay is a continuous process — the rate of decay is proportional to the current mass.

Final answer: 121.3 g

Diagram.



The exponential function plotted with key points marked.

Exponentials 58 (e11-058)

Difficulty: Foundation / Marks: 2

Solve $e^x = 5$. Give your answer in terms of $\ln$.

Worked solution.

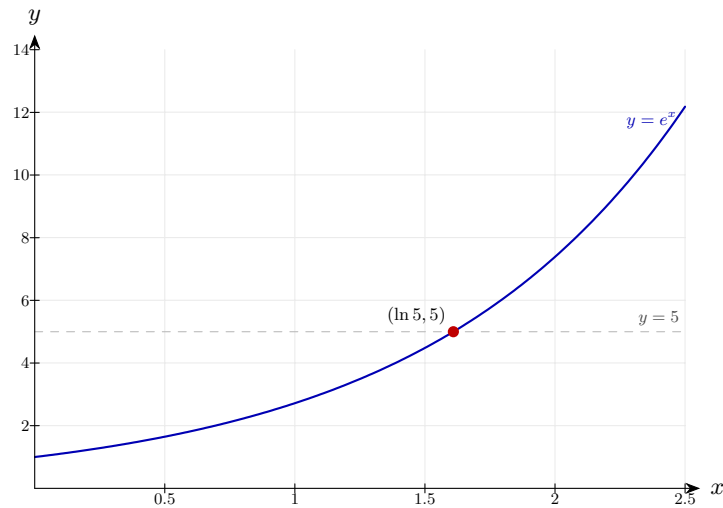
Take $\ln$ of both sides

$$x = \ln 5$$

The natural log $\ln$ is the inverse of e^x , so $\ln(e^x) = x$. Taking $\ln$ of both sides of $e^x = 5$ gives $x = \ln 5$ — leave this exact rather than using ≈ 1.609 .

Final answer: $x = \ln 5$

Diagram.



The exponential function plotted with key points marked.

Exponentials 59 (e11-059)

Difficulty: Foundation / *Marks: 3*

Solve $e^{2x} = 7$.

Worked solution.

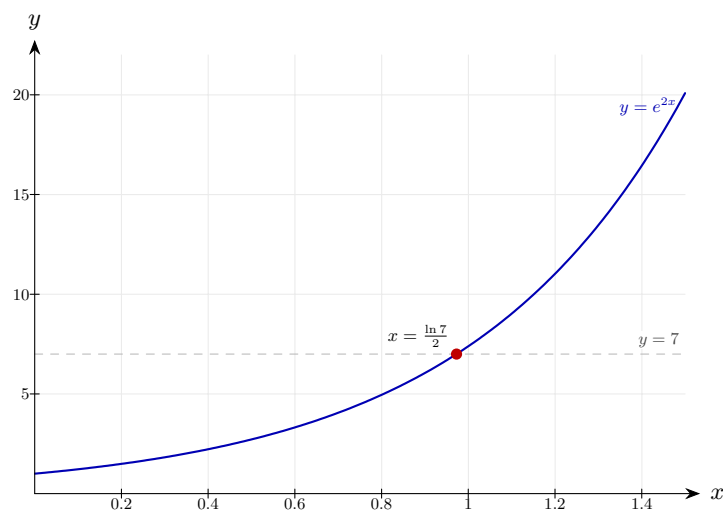
Take $\ln$

$$2x = \ln 7 \implies x = \frac{\ln 7}{2}$$

Applying $\ln$ to both sides uses $\ln(e^{2x}) = 2x$. Then divide by 2 to isolate x . Pitfall: do not split $\ln 7$ into $\ln 3 + \ln 4$ etc — it is already in its simplest exact form.

Final answer: $x = \frac{\ln 7}{2}$

Diagram.



The exponential function plotted with key points marked.

Exponentials 60 (e11-060)

Difficulty: Foundation / Marks: 3

Solve $3e^x - 6 = 0$.

Worked solution.

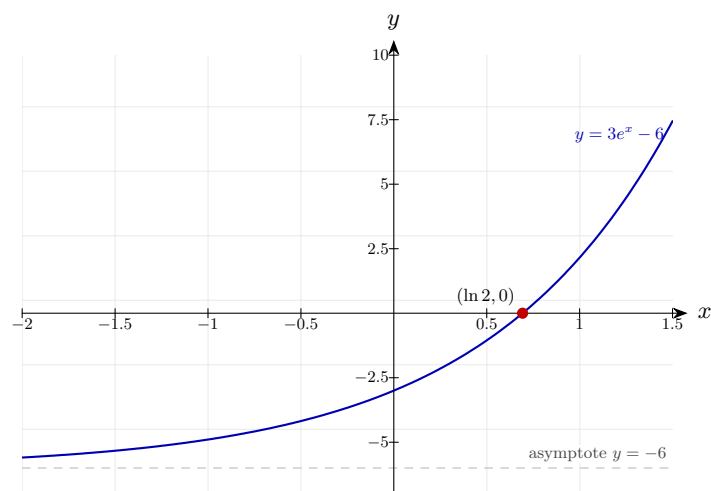
Rearrange

$$e^x = 2 \implies x = \ln 2$$

Isolate e^x first: add 6 to both sides and divide by 3 to get $e^x = 2$. Then take $\ln$ of both sides, using $\ln(e^x) = x$, to obtain $x = \ln 2$ (an exact answer).

Final answer: $x = \ln 2$

Diagram.



The exponential function plotted with key points marked.

Exponentials 61 (e11-061)

Difficulty: Foundation / Marks: 2

Evaluate $27^{\frac{1}{3}} + 16^{\frac{1}{4}}$.

Worked solution.

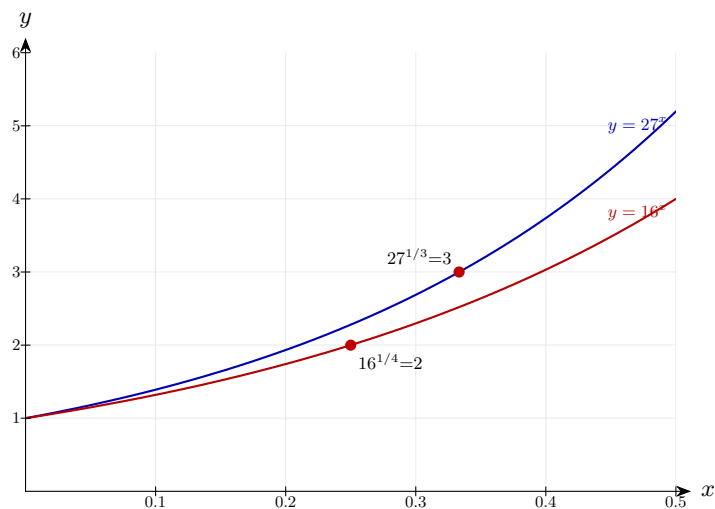
Evaluate each

$$3 + 2 = 5$$

Cube root of 27 is 3; fourth root of 16 is 2.

Final answer: 5

Diagram.



The exponential function plotted with key points marked.

Exponentials 62 (e11-062)

Difficulty: Foundation / Marks: 1

Simplify $(2^3)^4$.

Worked solution.

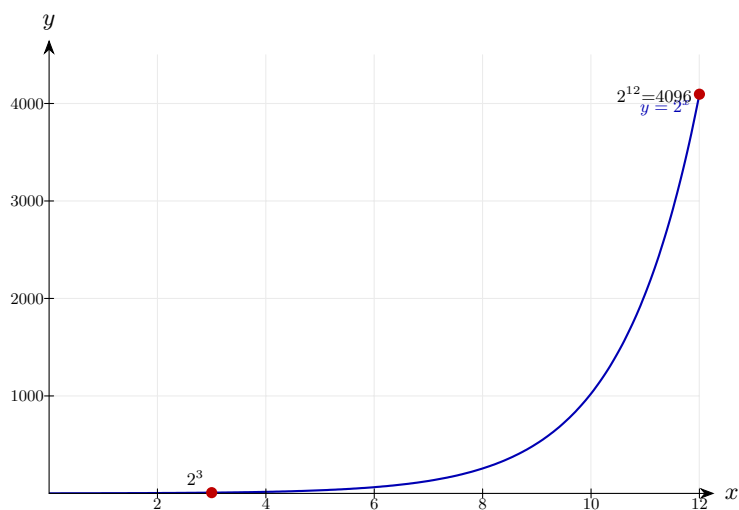
Multiply indices

$$2^{12} = 4096$$

When raising a power to a power, multiply the indices: $(a^m)^n = a^{mn}$. So $(2^3)^4 = 2^{3 \times 4} = 2^{12}$. Pitfall: do not add the indices — that rule is for multiplying powers with the same base, not nesting them.

Final answer: $2^{12} = 4096$

Diagram.



The exponential function plotted with key points marked.

Exponentials 63 (e11-063)

Difficulty: Foundation / Marks: 4

The graph of $y = a \times b^x$ passes through $(0, 3)$ and $(2, 48)$. Find a and b .

Worked solution.

Step 1: At $x = 0$

$$a \times b^0 = 3 \implies a = 3$$

Any base to the power 0 is 1, so the y -intercept of $y = ab^x$ is just a . The point $(0, 3)$ therefore pins down a immediately.

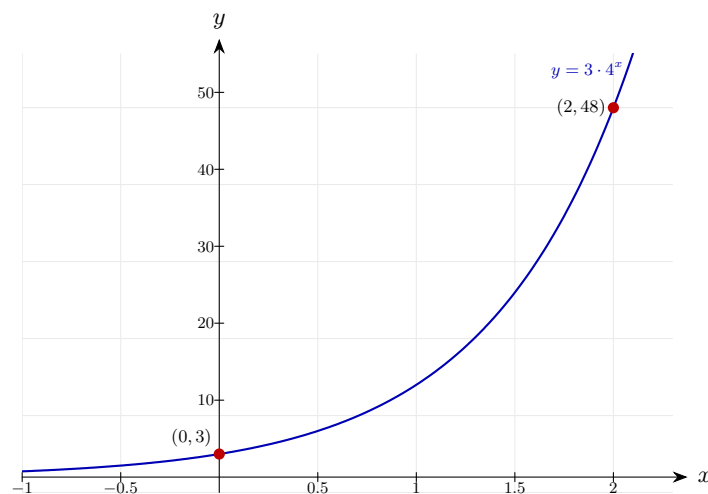
Step 2: At $x = 2$

$$3b^2 = 48 \implies b^2 = 16 \implies b = 4$$

Positive since exponential.

Final answer: $a = 3, b = 4$

Diagram.



The exponential function plotted with key points marked.

Exponentials 64 (e11-064)

Difficulty: Foundation / Marks: 2

Solve $2^{x+1} = 32$.

Worked solution.

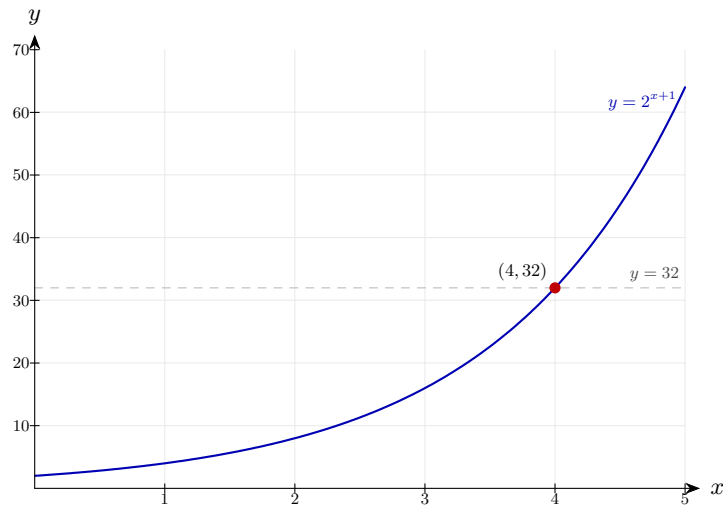
Write 32 as power of 2

$$2^{x+1} = 2^5 \implies x + 1 = 5 \implies x = 4$$

Express $32 = 2^5$ so both sides share base 2, then equate the exponents. Pitfall: the exponent is the entire expression $x + 1$, so subtract 1 (not 5) when solving for x .

Final answer: $x = 4$

Diagram.



The exponential function plotted with key points marked.

Exponentials 65 (e11-065)

Difficulty: Foundation / Marks: 5

The temperature T of a cooling object is modelled by $T = 20 + 80e^{-0.05t}$. Find: (a) the initial temperature; (b) the temperature after 20 minutes; (c) the long-term temperature.

Worked solution.

Step 1: (a) $t = 0$

$$T = 20 + 80 = 100^\circ$$

At $t = 0$, $e^{-0.05 \times 0} = e^0 = 1$, so the model collapses to $T = 20 + 80$. This is the starting temperature of the object.

Step 2: (b) $t = 20$

$$T = 20 + 80e^{-1} = 20 + 29.4 = 49.4^\circ$$

Substitute $t = 20$: the exponent becomes $-0.05 \times 20 = -1$, and $e^{-1} = \frac{1}{e} \approx 0.3679$, giving $80 \times 0.3679 \approx 29.4$. The natural base e arises because cooling is a continuous process (Newton's Law of Cooling).

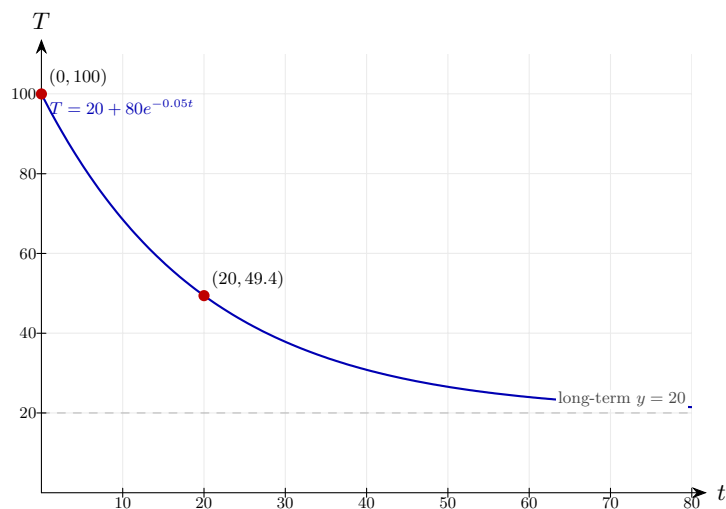
Step 3: (c) As $t \rightarrow \text{infinity}$

$$e^{-0.05t} \rightarrow 0 \implies T \rightarrow 20^\circ$$

Room temperature.

Final answer: (a) 100 degrees (b) 49.4 degrees (c) 20 degrees

Diagram.



The exponential function plotted with key points marked.

Exponentials 66 (e11-066)

Difficulty: Foundation / Marks: 2

Simplify $\frac{e^{3x} \times e^{2x}}{e^{4x}}$.

Worked solution.

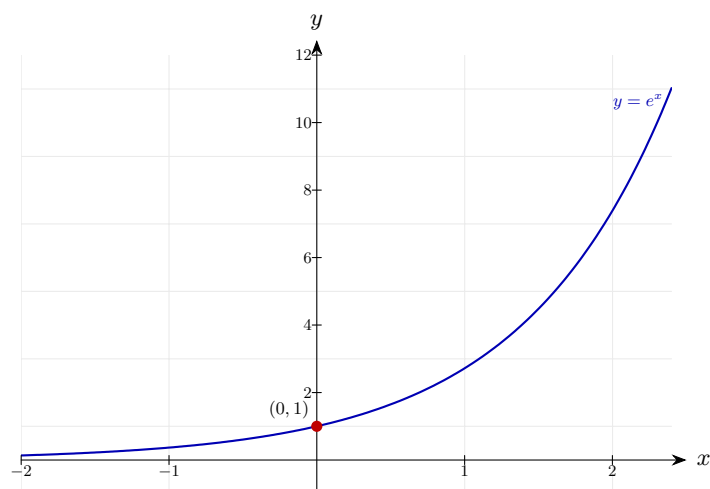
Index laws

$$\frac{e^{5x}}{e^{4x}} = e^x$$

First combine the numerator using $a^m \times a^n = a^{m+n}$: $e^{3x} \times e^{2x} = e^{5x}$. Then divide using $\frac{a^m}{a^n} = a^{m-n}$: $e^{5x-4x} = e^x$.

Final answer: e^x

Diagram.



The exponential function plotted with key points marked.

Exponentials 67 (e11-067)

Difficulty: Foundation / Marks: 5

Solve $e^{2x} - 5e^x + 6 = 0$.

Worked solution.

Step 1: Let $u = e^x$

$$u^2 - 5u + 6 = 0 \implies (u - 2)(u - 3) = 0$$

Notice $e^{2x} = (e^x)^2$, so the substitution $u = e^x$ turns the equation into a standard quadratic in u . Factorise to find the values of u .

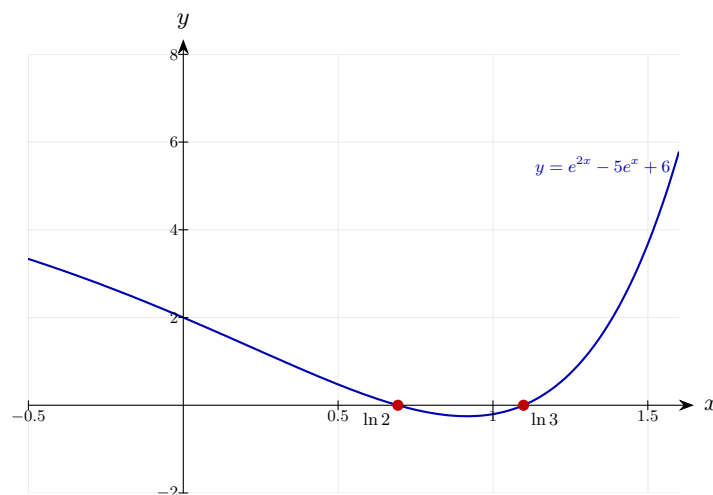
Step 2: Solve

$$e^x = 2 \implies x = \ln 2; \quad e^x = 3 \implies x = \ln 3$$

Reverse the substitution: each root of the quadratic gives an equation in x which is solved by taking $\ln$ of both sides. Both roots are valid because $u = e^x > 0$ for all real x , so neither root is rejected.

Final answer: $x = \ln 2$ or $x = \ln 3$

Diagram.



The exponential function plotted with key points marked.

Exponentials 68 (e11-068)

Difficulty: Foundation / Marks: 5

Solve $e^{2x} - 7e^x + 10 = 0$.

Worked solution.

Step 1: Let $u = e^x$

$$(u - 2)(u - 5) = 0 \implies u = 2 \text{ or } u = 5$$

Substituting $u = e^x$ (so $e^{2x} = u^2$) gives the quadratic $u^2 - 7u + 10 = 0$, which factorises to $(u - 2)(u - 5) = 0$.

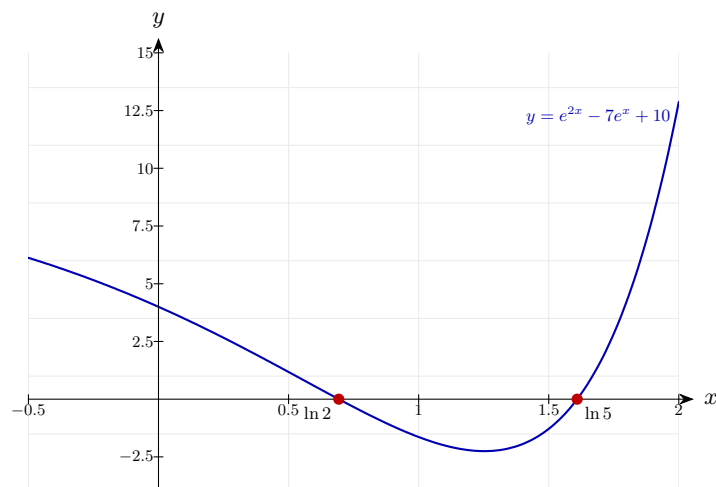
Step 2: Solve

$$x = \ln 2 \text{ or } x = \ln 5$$

Take $\ln$ of each value of u using $\ln(e^x) = x$. Both values of u are positive so both yield valid real x .

Final answer: $x = \ln 2$ or $x = \ln 5$

Diagram.



The exponential function plotted with key points marked.

Exponentials 69 (e11-069)

Difficulty: Foundation / Marks: 3

Write $y = 3 \times 2^x$ in the form $y = ae^{kx}$.

Worked solution.

Step 1: Rewrite 2^x

$$2^x = e^{x \ln 2}$$

Since $2 = e^{\ln 2}$.

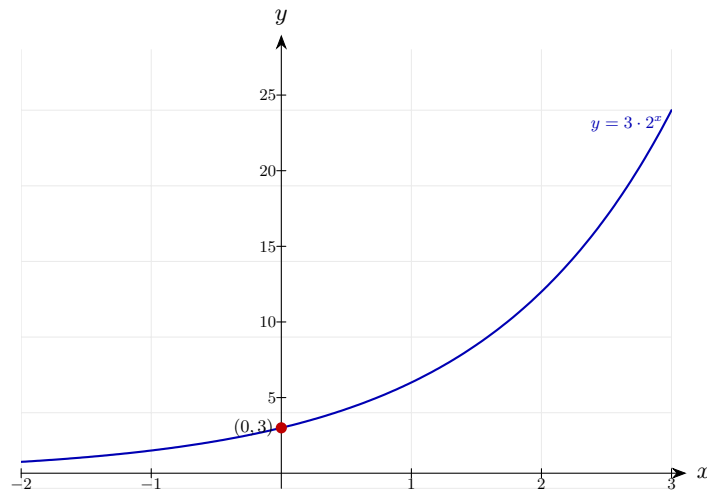
Step 2: Combine

$$y = 3e^{(\ln 2)x}$$

So $a = 3$, $k = \ln 2$.

Final answer: $y = 3e^{(\ln 2)x}$; $a = 3$, $k = \ln 2$

Diagram.



The exponential function plotted with key points marked.

Exponentials 70 (e11-070)

Difficulty: Foundation / Marks: 5

A car depreciates in value according to $V = 15000 \times 0.85^t$ where t is years. (a) Find the value after 3 years. (b) Find when the value drops below 5000.

Worked solution.

Step 1: (a) Substitute $t = 3$

$$V = 15000 \times 0.85^3 = 15000 \times 0.6141 = 9211$$

With a depreciation rate of 15% per year, the value is multiplied by 0.85 annually. After 3 years the multiplier is $0.85^3 \approx 0.6141$.

Step 2: (b) Solve

$$0.85^t = \frac{5000}{15000} = \frac{1}{3}$$

Divide both sides of $15000 \times 0.85^t = 5000$ by 15000 to isolate the exponential. This is the standard first step before taking logs.

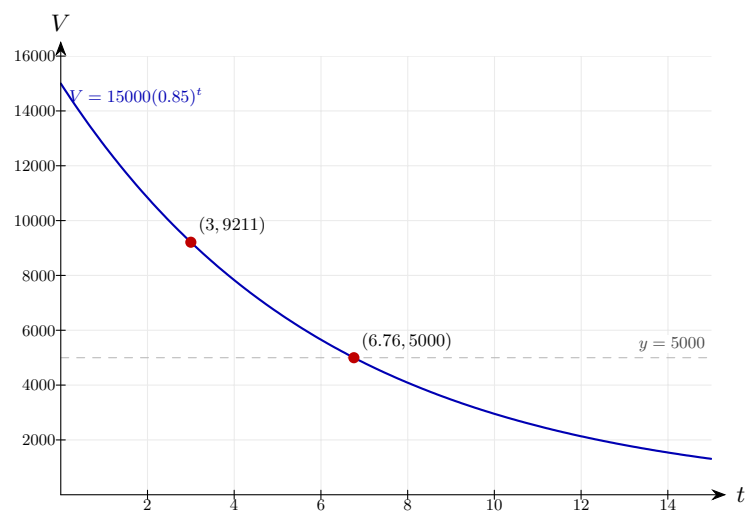
Step 3: Take logs

$$t = \frac{\ln(1/3)}{\ln 0.85} = \frac{-1.099}{-0.1625} = 6.76$$

Take $\ln$ of both sides; the rule $\ln(a^t) = t \ln a$ brings t down, giving $t \ln 0.85 = \ln \frac{1}{3}$. Both logs are negative (since their arguments are between 0 and 1), and the negatives cancel when dividing — so t is positive, as expected.

Final answer: (a) 9211 pounds (b) After approximately 6.76 years

Diagram.



The exponential function plotted with key points marked.
